



Media Engineering and Technology Faculty
German University in Cairo

Fundus Glaucoma Detection using Deep Learning

Bachelor Thesis

Author: Omar Walid Fathy Elsaid
Supervisors: Dr. Mahmoud Khalil
Submission Date: 26 May 2025



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This is to certify that:

- (i) the thesis comprises only my original work toward the Bachelor's Degree
- (ii) due acknowledgment has been made in the text to all other material used

Omar Walid Fathy Elsaid
26 May 2025

Abstract

Glaucoma, a leading cause of irreversible blindness worldwide, demands early and precise diagnosis to prevent vision loss. Traditional diagnostic approaches are often limited by high equipment costs and dependency on expert interpretation. This study explores the potential of deep learning techniques for automated glaucoma detection using fundus images from the SMDG-19 dataset, specifically captured using TOPCON fundus cameras. We implemented and evaluated three advanced convolutional neural network (CNN) architectures—ResNet50, EfficientNetV2-S, and MobileNetV3—for binary classification of fundus images as either glaucomatous or normal.

Our methodology involved a comprehensive analysis and deep understanding of each model’s architecture, including the functional role of individual layers and their contributions to feature extraction. Preprocessing steps such as CLAHE (Contrast Limited Adaptive Histogram Equalization), gamma correction, and random data augmentation were applied to enhance model robustness and generalizability. We also employed transfer learning by initializing the models with ImageNet pre-trained weights, followed by fine-tuning for glaucoma detection. The models were assessed using multiple performance metrics, including accuracy, precision, recall, F1 score, and AUC, achieving outstanding classification results, with accuracies ranging from 98% to 99.2% and near-perfect AUC values. These findings highlight the promise of deep learning-based systems as effective, scalable, and accessible tools for early glaucoma screening.

The experimental results demonstrated consistently high performance across the three deep learning models—ResNet50, EfficientNetV2-S, and MobileNetV3—with classification accuracies ranging from 98% to 99.2%. This performance was achieved through rigorous preprocessing, including image normalization, CLAHE, gamma correction, and randomized data augmentation, all of which contributed to enhancing the models’ robustness and ensuring fair evaluation conditions. A thorough analysis was conducted on the SMDG-19 dataset, which contains high-quality fundus images captured using TOPCON cameras, enabling effective identification of glaucomatous features. Additionally, detailed examination of each model’s configuration and architecture was performed to understand the impact of individual layers on learning performance. The models also achieved near-perfect AUC values, further validating their ability to distinguish between glaucomatous and normal fundus images with high reliability.

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Chapter 1

Introduction

1.1 Background

Glaucoma is a severe optic nerve disease and the second leading cause of irreversible blindness worldwide, often progressing silently without noticeable symptoms until significant vision loss occurs. It primarily results from elevated intraocular pressure (IOP), which damages the nerve fibers responsible for transmitting visual information to the brain. It is the second leading cause of permanent blindness globally, with an estimated 95 million individuals affected, and approximately 10 million experiencing blindness in at least one eye [22]. Early detection is critical but remains challenging, particularly in resource-limited regions such as many African countries, due to financial constraints, limited access to specialized diagnostic equipment, and shortages of trained professionals.

Traditionally, diagnosing glaucoma has relied on Optical Coherence Tomography (OCT), a highly effective but expensive imaging technique requiring specialized expertise. The development of affordable portable fundus cameras, coupled with advancements in deep learning—particularly Convolutional Neural Networks (CNNs)—offers a promising solution for accessible, cost-effective glaucoma screening.

One of the key structural indicators of glaucoma is the optic cup-to-disc ratio (CDR). In healthy eyes, the optic cup (the central bright region of the optic disc) remains relatively small compared to the entire optic disc. In glaucomatous eyes, the cup enlarges, leading to an increased CDR. However, relying solely on CDR can be insufficient, as some individuals naturally have larger CDRs (>0.6) without glaucoma. Therefore, modern clinical evaluation also considers additional features such as neuroretinal rim thinning, localized notching, peripapillary atrophy, retinal nerve fiber layer (RNFL) defects, asymmetry between the two eyes, vessel bending (bayonetting), and characteristic patterns of visual field loss.

1.1.1 The Annual Economic Burden

The economic burden of glaucoma is substantial. In Australia, the economic impact is projected to exceed AUD 4.3 billion by 2025 [23]. In the United States, the annual economic burden is approximately USD 2.9 billion [24]. In Europe, the average annual direct medical cost per glaucoma patient ranges from €429 to €523, with total annual costs, including indirect expenses, estimated between €11,758 and €19,111 [26]. In developing countries like Nigeria, patients may spend up to 10% of their monthly income on glaucoma care, highlighting the financial strain imposed by the disease [25].

1.2 Current Challenges in Glaucoma Diagnosis

Diagnosing glaucoma presents significant challenges due to its asymptomatic early stages, heterogeneous presentation, and limitations in current diagnostic methods. In the early phases, patients typically experience no noticeable symptoms, and peripheral vision loss often goes undetected until irreversible damage occurs. Reliance on intraocular pressure (IOP) measurements is problematic because up to 40% of glaucoma cases (e.g., normal-tension glaucoma) occur with normal IOP levels. Additionally, subjective assessments of the optic nerve head and retinal nerve fiber layer can vary widely among clinicians, leading to inconsistencies in detection. Advanced tools like optical coherence tomography (OCT) and visual field testing improve accuracy but are often inaccessible in low-resource settings, contributing to underdiagnosis, particularly in developing countries. Furthermore, slow disease progression means patients may not seek timely screenings, while mimicking conditions (e.g., ischemic optic neuropathy) can lead to misdiagnosis. These barriers collectively result in 50% of glaucoma cases remaining undetected globally, emphasizing the need for better screening protocols, affordable diagnostics, and increased awareness.

1.3 Research Objectives

This study aims to :

- To implement and evaluate deep learning models, specifically ResNet50, EfficientNetV2-S, and MobileNetV3, for binary classification of fundus images in glaucoma detection.
- To assess the feasibility of deploying a web-based application for automated glaucoma assessment and diagnosis support.
- To analyze and understand the architectures of deep learning models and explore possible transformations and adjustments to enhance model accuracy in detecting glaucoma.

1.4 Scope and Limitations

Scope

- Research and identification of the most effective classification models and datasets for glaucoma detection using deep learning.
- In-depth analysis of classification model architectures, including the theoretical role of layers and the impact of different optimizers.
- Exploration of model adjustments and preprocessing techniques to enhance detection accuracy.
- Comparative evaluation of model performance (EfficientNetV2-S, ResNet50, MobileNetV3) in terms of accuracy and practical deployment suitability.

Limitations

This project encountered several limitations. First, glaucoma is a complex disease affecting multiple regions of the optic nerve, making object detection models like YOLO unsuitable for this

task. Dataset limitations were also significant: most available ophthalmic datasets are small, lack domain diversity, and do not fully represent different demographics, potentially leading to biased results. For instance, individuals with darker skin tones may exhibit anatomical features that result in overdiagnosis, while underdiagnosis may occur in other populations. Moreover, models were trained solely on color fundus images without multimodal inputs such as OCT scans or visual field tests, which restricted the comprehensiveness of diagnosis. Variability in clinical labeling, the asymptomatic nature of early glaucoma, and patient non-compliance further complicated model training and evaluation. These challenges emphasize the need for more diverse, multi-source datasets and integrative diagnostic approaches in future research.

1.5 Thesis Structure

Thesis Organization

The remainder of this thesis is organized as follows:

- **Chapter 2: Literature Survey** — Reviews existing research related to glaucoma detection, deep learning techniques in ophthalmology, and the challenges associated with early diagnosis.
- **Chapter 3: Methodology** — Describes the datasets, preprocessing methods, model architectures (ResNet50, EfficientNetV2-S, MobileNetV3), training strategies, and evaluation procedures used in this study.
- **Chapter 4: Results and Discussion** — Presents the experimental results and compares model performances.
- **Chapter 5: Conclusion and Future Work** — Summarizes the key contributions of the research and outlines possible directions for future enhancements in automated glaucoma screening systems.

Chapter 2

Literature Review

2.1 The Available Datasets

2.1.1 Databases Used

Several publicly available and proprietary datasets are used for training and evaluating deep learning models for glaucoma detection:

- **Eye_Diseases_Classification** – <https://www.kaggle.com/datasets/gunavenkatdoddi/eye-diseases-classification>
- **ORIGA Dataset** – <https://www.kaggle.com/datasets/arnavjain1/glaucoma-datasets/data>
- **Drishti-GS** – <https://www.kaggle.com/datasets/lokeshsaipureddi/drishtigs-retina-datas>
- **Refugee** – <https://www.kaggle.com/datasets/arnavjain1/glaucoma-datasets/data>
- **Rim-One** – <https://www.kaggle.com/datasets/lucascunhadecarvalho/rimoner2>
- **SMDG-19** – <https://www.kaggle.com/datasets/deathtrooper/multichannel-glaucoma-bench>
- **G1020** – <https://www.kaggle.com/datasets/arnavjain1/glaucoma-datasets>

2.1.2 Eye_Diseases_Classification Dataset

The Eye Diseases Classification Dataset is a publicly available dataset on Kaggle, created for developing deep learning models to classify various eye diseases. It consists of color fundus images that can be used for multi-class classification tasks. The dataset is particularly useful for researchers in the field of medical image analysis, especially for ophthalmology-focused AI applications. It provides labeled fundus images of different eye conditions, making it suitable for training convolutional neural networks (CNNs) and other deep learning models for disease detection.

Source Datasets and Composition

- **IDRiD (Indian Diabetic Retinopathy Image Dataset)**
- **Ocular Recognition Dataset**
- **HRF (High-Resolution Fundus Image Dataset)**

Class Distribution

The dataset includes images categorized into multiple eye disease classes, though the exact number of images per class is not always explicitly mentioned. Some common classes in eye disease classification datasets typically include:

- **Normal** (25%)
- **Diabetic Retinopathy** (25%)
- **Glaucoma** (25%)
- **Cataract** (25%)

Image Properties

In the dataset, glaucoma images have a resolution of 256×256 pixels, whereas images in the Normal class are of a higher resolution, measuring 512×512 pixels.

Dataset Samples



Figure 2.1: Glaucoma Sample



Figure 2.2: Normal Eye

Figure 2.3: Comparison of Glaucoma and Normal Eye Images

2.1.3 ORIGA Dataset

The ORIGA (Optic Nerve Head-based Glaucoma) Dataset is a widely used fundus image dataset for glaucoma detection and segmentation tasks. It contains annotated fundus images with glaucoma and non-glaucoma labels, making it valuable for training machine learning models. The dataset includes optic disc (OD) and optic cup (OC) segmentations, crucial for glaucoma assessment.

Class Distribution

The ORIGA (Optic Nerve Head-based Glaucoma) Dataset is a widely used fundus image dataset for glaucoma detection and segmentation tasks. It contains annotated fundus images with glaucoma and non-glaucoma labels, making it valuable for training machine learning models. The dataset includes optic disc (OD) and optic cup (OC) segmentations, crucial for glaucoma assessment.

- **Total Images:** Approximately 650 fundus images
- **Glaucoma Cases:** Around 168 images , 26%
- **Non-Glaucoma Cases:** Around 482 , 74% images

Image Properties

In the dataset, glaucoma images have a width of 2048 pixels and the height ranges from 2480-2546.

Dataset Samples

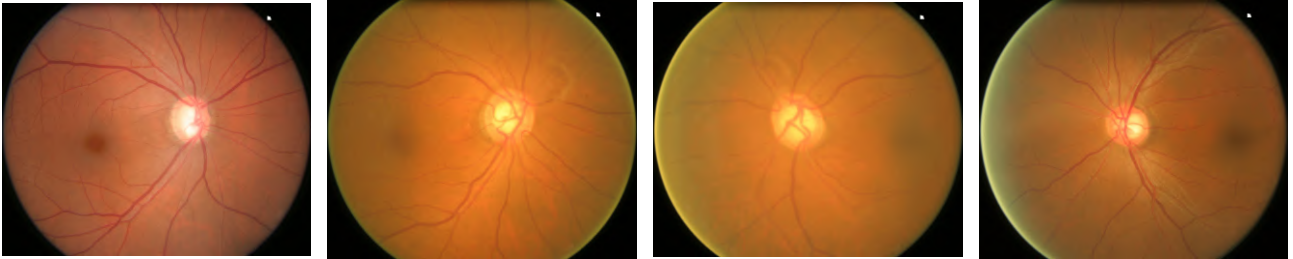


Figure 2.4: OD Glaucoma Sample

Figure 2.5: OD Normal Eye

Figure 2.6: OS Glaucoma Eye

Figure 2.7: OS Normal Eye

Figure 2.8: Comparison of Glaucoma and Normal Eye Images in the ORIGA Dataset.

2.1.4 Drishti-GS

The DRISHTI-GS dataset, hosted on Kaggle, is a specialized resource for optic nerve head (ONH) segmentation and glaucoma detection. Collected by the Aravind Eye Hospital in India, it includes retinal fundus images with ground truth annotations for optic disc and cup boundaries. This dataset is widely used for developing and evaluating algorithms for glaucoma diagnosis, optic disc segmentation, and calculating the cup-to-disc ratio (CDR), making it a valuable tool for advancing research in ophthalmology and computer-aided diagnosis.

Class Distribution

- **Total Images:** Approximately 101 fundus images
- **Glaucoma Cases:** Around 70 images
- **Non-Glaucoma Cases:** Around 31 images

Image Properties:

Image Format

- The images are stored in standard formats such as **.png**

Resolution

- The images have a high resolution of **2049×1755 pixels**.
- This high resolution allows for detailed analysis of the optic disc, optic cup, and other retinal structures.
- The images are in RGB format (3 channels: Red, Green, Blue).

Dataset Samples



Figure 2.9: Glaucoma Eye

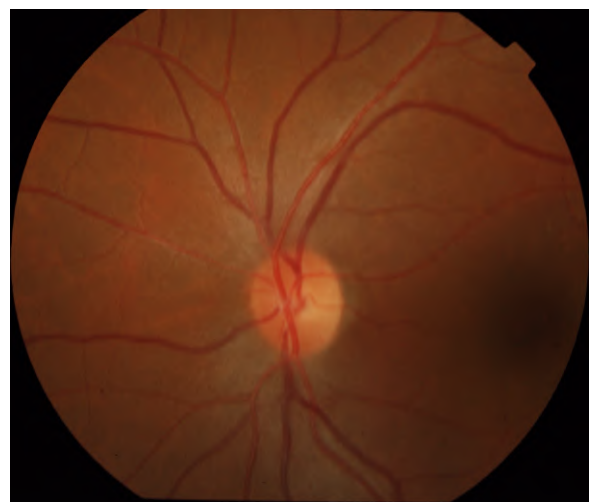


Figure 2.10: Normal Eye

Figure 2.11: Comparison of Glaucoma and Normal Eye Images in the DRISHTI-GS Dataset.

2.1.5 G1020

The G1020 dataset is a publicly available collection of retinal fundus images aimed at assisting the development and evaluation of glaucoma detection models. The images in this dataset have been sourced from multiple repositories and clinical settings, and they contain annotations for optic disc and optic cup segmentation. This dataset is often used for training and evaluating machine learning algorithms in ophthalmology and medical image analysis.

Class Distribution

- **Total Images:** Approximately 1020 fundus images
- **Glaucoma Cases:** Around 724 images, 71%
- **Non-Glaucoma Cases:** Around 296 images, 29%

Image Properties

- Image width by height **3004×2423**
- **RGB** color Format
- The images are stored in standard formats such as **.JPG**

Preprocessed Data Organization

The G1020 dataset includes preprocessed versions of the original fundus images to enhance uniformity and ease of processing:

- **Nerve-Removed Images:** A folder containing images where the optic nerve has been removed, focusing on relevant retinal structures.
- **Descaled Images:** Another folder where all images have been resized to **512 × 512 pixels**, standardizing dimensions across the dataset for consistency.
- These preprocessing steps help in simplifying image processing tasks, ensuring compatibility across deep learning models.

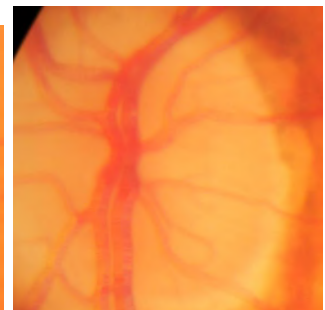
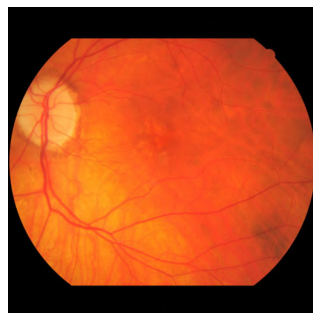
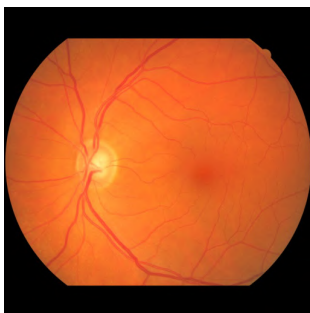


Figure 2.12: Nerve Present Normal Eye

Figure 2.13: Nerve Present Glaucoma Eye

Figure 2.14: Nerve Cropped Normal Eye

Figure 2.15: Nerve Cropped Glaucoma sample

Figure 2.16: Comparison of Glaucoma and Normal Eye Images in the G1020 Dataset.

2.1.6 REFUGE Dataset Analysis

The **REFUGE (Retinal Fundus Glaucoma Challenge) dataset** is a widely used dataset for glaucoma detection and optic disc/cup segmentation in fundus images. Below is an analysis of its key components:

Dataset Overview

- The dataset consists of high-resolution **fundus images** labeled for **glaucoma diagnosis** and **optic disc segmentation**.
- It is commonly used for automated **glaucoma detection**, **optic disc (OD) segmentation**, and **optic cup (OC) classification**.
- The dataset Guide is stored in **JSON file** format

Image Properties

- The fundus images have **high resolution RGB**, allowing precise segmentation and analysis **2124×2056**.
- image format is **JPG**
- **optic disc and optic cup**, making it useful for supervised learning.

Class Distribution

- The dataset is divided into glaucoma-positive with (1080) about **(90%)** and glaucoma-negative with (120) images about **(10%)**.
- It includes ground truth segmentation masks for both the **optic disc** and **optic cup**, assisting in developing deep learning models.

Preprocessing and Data Augmentation

- The dataset may include preprocessed versions, such as **cropped** and **resized** images, to standardize input sizes across models.

Dataset Samples

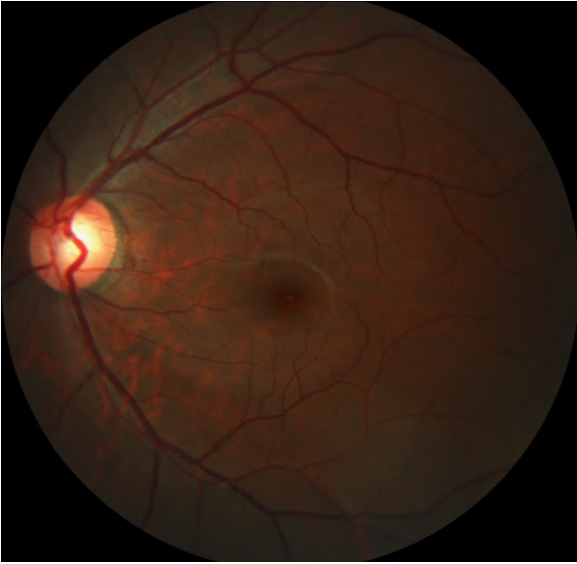


Figure 2.17: Glaucoma Eye

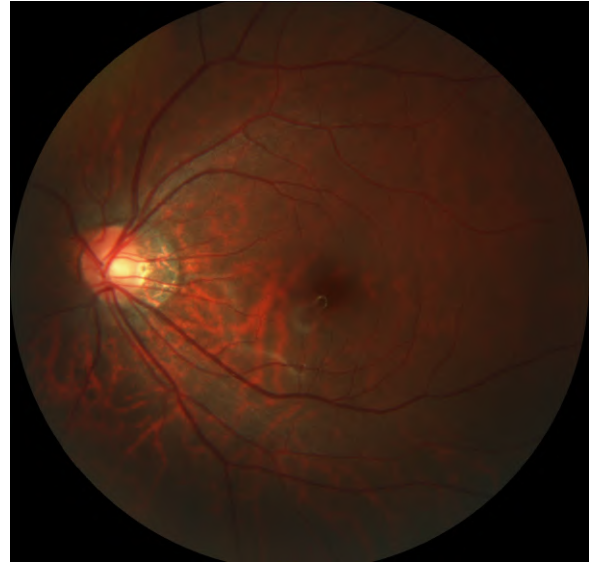


Figure 2.18: Normal Eye

Figure 2.19: Comparison of Glaucoma and Normal Eye Images in the REFUGE Dataset.

2.1.7 SMDG-19

The SMDG (Standardized Multi-Channel Dataset for Glaucoma) is a benchmark dataset for glaucoma detection and optic disc segmentation. It is composed of fundus images from 19 publicly available datasets, providing a diverse and well-curated dataset for researchers in the field of ophthalmology and deep learning. The dataset includes multiple image channels and annotations to aid in robust machine learning model training.

Source Datasets and Composition

The SMDG dataset is composed of images collected from multiple publicly available datasets. Below is a list of the datasets used:

- **BEH (Bangladesh Eye Hospital)**
- **CRFO-v4**
- **DR-HAGIS (Diabetic Retinopathy, Hypertension, Age-related Macular Degeneration, and Glaucoma ImageS)**
- **DRISHTI-GS1-TRAIN**
- **DRISHTI-GS1-TEST**
- **EyePACS-AIROGS**
- **FIVES**
- **G1020**
- **HRF (High Resolution Fundus)**
- **JSIEC-1000 (Joint Shantou International Eye Center)**
- **LES-AV**

- OIA-ODIR-TRAIN
- OIA-ODIR-TEST-ONLINE
- OIA-ODIR-TEST-OFFLINE
- ORIGA-light
- PAPILA
- REFUGE1-TRAIN (Retinal Fundus Glaucoma Challenge 1 Train)
- REFUGE1-VALIDATION (Retinal Fundus Glaucoma Challenge 1 Validation)
- sjchoi86-HRF

2.1.8 Image Properties

The SMDG dataset is composed of fundus images from 19 different sources, resulting in diverse image characteristics. Below are the key properties of the dataset:

- **Color Variations:**
 - The dataset includes both **RGB (colored)** and **grayscale (black and white)** images.
 - The **BEH (Bangladesh Eye Hospital)** dataset contains black and white images, while most datasets are in RGB format.
 - Differences in camera sensors and acquisition methods introduce slight color tone variations across different sources.
- **Resolution Differences:**
 - The images originate from multiple datasets, each with **different resolutions**.
 - A **preprocessed version** of the dataset resizes all images to **512×512 pixels** to ensure uniformity.
- **Lighting and Contrast Variability:**
 - Different imaging conditions lead to **overexposed and underexposed images**.
 - Some images have **high contrast**, while others appear washed out, affecting feature visibility.
- **Angle and View Differences:**
 - Fundus images are captured from **different angles and zoom levels**, leading to variations in optic disc and cup visibility.
 - Some datasets include **wide-field and close-up views** of the optic nerve head, influencing the field of view.
- **Preprocessing and Standardization:**
 - A separate set of images is **descaled to 512×512 pixels**, making them easier to process in deep learning models.

Dataset Samples

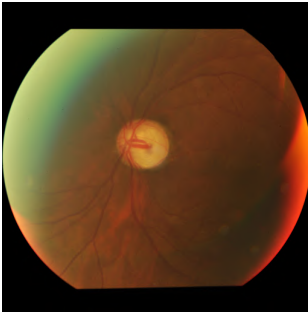


Figure 2.20: BEH Dataset

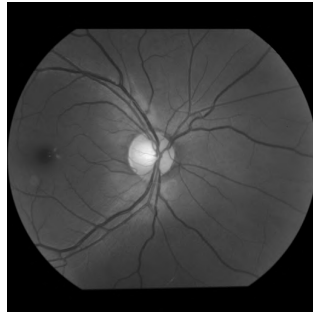


Figure 2.21: BEH Dataset

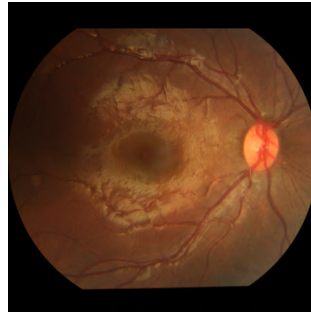


Figure 2.22: BEH Dataset

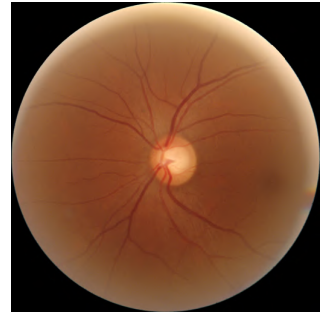


Figure 2.23: CRFO-v4 Dataset



Figure 2.24: CRFO-v4 Dataset

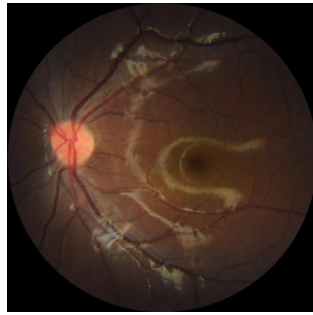


Figure 2.25: FIVES Dataset

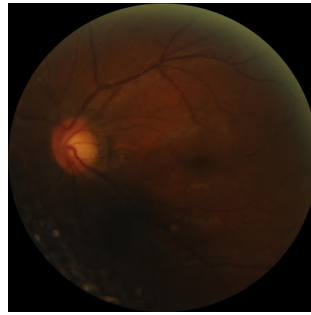


Figure 2.26: OIA-ODIR TEST Dataset



Figure 2.27: OIA-ODIR TEST Dataset

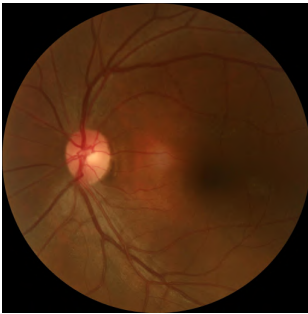


Figure 2.28: OIA-ODIR TEST Dataset

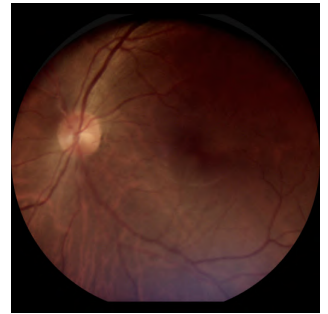


Figure 2.29: OIA-ODIR TEST Dataset

Figure 2.30: Sample images from different datasets in SMDG dataset.

2.1.9 Dataset Overview

Dataset	Normal	Glaucomatous	Resolution	Total
eye_diseases_classification	1074	1007	512×512 JPG	2081
ORIGA	482	168	2513×2048 JPG	650
Drishti_GS1	31	70	2049×1755 PNG	101
G1020	296	724	2124×2056 JPG	1020
REFUGEE	120	1080	3965×3000 PNG	1200
SMDG-19	10,586	6,656	512×512 JPG	17242
Total	11,530	8,713	-	20,243

Table 2.1: Fundus Glaucoma Detection Datasets

2.1.10 Conclusion

The most reputable and widely used fundus glaucoma datasets, including ORIGA, REFUGE, G1020, and the SMDG dataset, provide valuable resources for glaucoma detection. Among these, SMDG stands out as the largest and most comprehensive dataset, making it particularly suitable for our project. These datasets often include augmented and preprocessed images, with optic nerve head (ONH) removal to facilitate analysis. Additionally, image resolutions are unified to enhance compatibility and streamline preprocessing workflows.

However, certain challenges arise when working with these datasets. In ORIGA and REFUGE, the imbalance in glaucoma and non-glaucoma cases introduces a potential bias during model training, which may negatively impact performance. Specifically, an unequal distribution of positive and negative cases may hinder the model’s ability to generalize effectively. In contrast, the SMDG dataset offers a more balanced ratio of glaucoma to non-glaucoma cases (**Normal: 10,586** and **Glaucoma: 6,656**), reducing the risk of model bias while maintaining a substantial dataset size.

Another significant challenge is the variation in image acquisition across different datasets. Due to differences in imaging protocols, some images appear in grayscale while others are in RGB, resulting in inconsistencies in color distribution and contrast. Additionally, variations in lighting conditions and image angles further complicate the uniformity of data preprocessing. These factors must be carefully managed through normalization techniques to ensure robust model performance.

In conclusion, while the SMDG dataset provides the most suitable foundation for glaucoma detection due to its size, preprocessing, and balanced distribution, the inherent variability across datasets necessitates careful preprocessing and augmentation strategies to ensure reliable model training and evaluation.

2.2 Success Metrics

The performance of deep learning models for glaucoma detection is typically evaluated using:

Metric	Description	Expression
Accuracy (ACC)	It is the number of correct predictions divided by the total number of predictions.	$ACC = \frac{TP+TN}{TP+TN+FP+FN}$
Top-1 Accuracy	It measures the percentage of cases where the model's highest confidence prediction matched the true label.	$Top-1 = \frac{\text{Correct Predictions (Highest Confidence)}}{\text{Total Samples}}$
Top-5 Accuracy	It measures the percentage of cases where the correct label appears within the model's top five predictions. Useful in multi-class problems.	$Top-5 = \frac{\text{Samples with True Label in Top 5}}{\text{Total Samples}}$
Loss	Binary classification uses binary cross-entropy loss. Multi-class classification uses categorical cross-entropy loss.	$l(y, p) = -(y \log p + (1 - y) \log(1 - p))$ $l(y, p) = -\sum_{i=1}^M y_i \log p_i$
F1 Score	Harmonic mean of precision and recall. Ranges from [0,1]. Higher F1 Score means better performance.	$F1 = \frac{2TP}{2TP+FP+FN}$
Recall	Correct positive results relative to all relevant samples.	$Recall = \frac{TP}{TP+FN}$
Precision	Correct positive results relative to all positive predictions.	$P = \frac{TP}{TP+FP}$
ROC-AUC	Plots True Positive Rate (TPR) against False Positive Rate (FPR) to determine an optimal threshold.	$TPR = \frac{TP}{TP+FN}, \quad FPR = \frac{FP}{FP+TN}$

Continued on next page

Continued from previous page

Metric	Description	Expression
Matthews Correlation Coefficient (MCC)	Measures correlation between true and predicted values. A value close to +1 indicates a strong positive correlation, 0 means no correlation, and -1 indicates total disagreement between predictions and actual labels.	$MCC = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}}$
Confusion Matrix	Summarizes the model's classification results by showing the number of true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). It helps in understanding model performance and deriving other key metrics.	TP: Correctly predicted positive cases. TN: Correctly predicted negative cases. FP: Incorrectly predicted positive cases (false alarms). FN: Incorrectly predicted negative cases (missed positives).

2.3 Results

This is a comparison of 9 tests with different hyper-parameters and different sizes of datasets

Ref. Authors (Year)	Architecture Used	Dataset	Classes	Classification Rate	Category
Proposed Model (2023)	13-layer CNN	HRF, Origa, Drishti, LF Hospital	Normal, Glaucomatous	93.86% (Soft-Max), 95.61% (SVM)	Deep Learning
Chen et al. (2016) [14]	6-layer CNN	ORIGA, SCES	Normal, Glaucomatous	AUC: 0.831 (ORIGA), 0.887 (SCES)	Deep Learning
Ahn et al. (2017) [2]	3-layer CNN	Private Dataset (1542 images)	Normal, Glaucomatous	87.9% Accuracy, AUC: 0.94	Deep Learning
Raghavendra et al. (2018) [3]	4-layer CNN	Private Dataset	Normal, Glaucomatous	98.13% Accuracy	Deep Learning
Baida Al Bander et al. (2018) [4]	AlexNet (23 layers) + SVM	RIMONE	Normal, Glaucomatous	88.2% Accuracy, 85% Sensitivity, 90.8% Specificity	Deep Learning + SVM
Blagus R et al. (2013) [10]	Multi-branch Neural Network	Private Dataset	Normal, Glaucomatous	91.51% Accuracy, 92.33% Sensitivity, 90.90% Specificity	Deep Learning
Mojab N et al. (2019) [11]	Deep CNN	Private Dataset	Normal, Glaucomatous	79.04% Accuracy, 79.04% Sensitivity, 88.95% Specificity	Deep Learning
Bajwa M N et al. (2019) [12]	Two-stage RCNN	Private Dataset	Normal, Glaucomatous	95.45% Accuracy, 96.43% Sensitivity, 93.45% Specificity	Deep Learning
Burlina P et al. (2016) [13]	SVM Classifier	Private Dataset	Normal, Glaucomatous	95.00% Accuracy, 96.40% Sensitivity, 95.60% Specificity	SVM

Ref. Authors (Year)	Architecture Used	Dataset	Classes	Classification Rate	Category
Tan J H et al. (2018) [1]	Deep CNN	Private Dataset	Normal, Glaucomatous	93.45% Accuracy, 79.04% Sensitivity, 96.43% Specificity	Deep Learning
Chai et al. (2018) [18]	RCNN	Private Dataset (2554 images)	Normal, Glaucomatous	91.51% Accuracy, 92.33% Sensitivity, 90.90% Specificity	Deep Learning
Proposed Model (2023)	ResNet-50	UZL, REFUGE	Glaucoma, Non-Glaucoma	AUC: 0.94 (UZL), 0.87 (REFUGE)	Deep Learning
Proposed Model (2023)	ResNet-50	UZL	VCDR Estimation	R ² : 77%, MAE: 0.079	Deep Learning
Proposed Model (2023)	ResNet-50 (ONH Crop 60%)	UZL	Glaucoma, Non-Glaucoma	AUC: 0.87 (UZL), 0.80 (REFUGE)	Deep Learning
Proposed Model (2023)	ResNet-50 (ONH Crop 60%)	UZL	VCDR Estimation	R ² : 37%, MAE: 0.142	Deep Learning
Proposed Model (2023)	ResNet-50 (Periphery Crop 30%)	UZL	Glaucoma, Non-Glaucoma	AUC: 0.93	Deep Learning
Proposed Model (2023)	ResNet-50 (Periphery Crop 30%)	UZL	VCDR Estimation	R ² : 77%, MAE: 0.078	Deep Learning
Proposed Model (2023)	ResNet-50 (Periphery Crop 1%)	UZL	Glaucoma, Non-Glaucoma	AUC: 0.67	Deep Learning
Proposed Model (2023)	ResNet-50 (Periphery Crop 1%)	UZL	VCDR Estimation	R ² : 0.01, MAE: 0.189	Deep Learning

Table 2.3: Summary of results from the research papers

2.4 Comparison of Methods and Techniques for Glaucoma Detection

Test 1: Proposed Model - CNN-SoftMax

This test employed a 13-layer Convolutional Neural Network (CNN) with Max Pooling and ReLU activation. The model used a SoftMax classifier for glaucoma detection. It achieved an accuracy of 93.86%, specificity of 100%, sensitivity of 85.42%, and an AUC of 92.71%. The high specificity indicates that the model produced no false positives for normal cases, making it highly reliable for glaucoma screening.

Test 2: Proposed Model - CNN-SVM

This test used the same 13-layer CNN architecture but replaced the SoftMax classifier with an SVM classifier. The model achieved an accuracy of 95.61%, specificity of 100%, sensitivity of 89.58%, and an AUC of 94.79%. The SVM classifier outperformed the SoftMax-based model in accuracy and sensitivity, demonstrating its effectiveness in glaucoma detection.

[14] Test 3: Chen et al. (2014)

A 6-layer CNN was implemented for glaucoma screening. The model was trained and evaluated on the ORIGA and SCES datasets, achieving AUC values of 83.1% and 88.7%, respectively. These results indicate a strong ability to distinguish between normal and glaucomatous eyes, though accuracy and sensitivity were not reported.

[16] Test 4: Ahn et al. (2017)

This test utilized a 3-layer CNN with Max Pooling applied after each convolutional layer, followed by two fully connected layers. The model was evaluated on a private dataset of 1,542 images, achieving an accuracy of 87.9% and an AUC of 94%. The results demonstrate the model's reliability in glaucoma detection.

[3] Test 5: Raghavendra et al. (2018)

A 4-layer CNN was employed, incorporating batch normalization, ReLU activation, and Max Pooling at each layer. The model achieved an accuracy of 98.13% on a private dataset, showcasing exceptional performance in glaucoma classification. However, specificity, sensitivity, and AUC were not reported.

[4] Test 6: Baida Al Bander et al. (2018)

This test used a pretrained AlexNet model (23 layers) with an SVM classifier for glaucoma detection. The model was evaluated on the RIMONE database, achieving an accuracy of 88.2%, specificity of 90.8%, and sensitivity of 85%. The results highlight the model's robustness in handling diverse datasets.

[10] Test 7: Blagus R et al. (2013)

A multi-branch neural network was employed, combining features from multiple layers for glaucoma classification. The model achieved an accuracy of 91.51%, specificity of 90.90%, and sensitivity of 92.33%, demonstrating its ability to capture complex patterns in fundus images.

[11] Test 8: Mojab N et al. (2019)

A deep CNN was used, trained on a dataset of fundus images. The model achieved an accuracy of 79.04%, specificity of 88.95%, and sensitivity of 79.04%. While the performance was moderate, the model showed potential for further improvement.

[12] Test 9: Bajwa M N et al. (2019)

This test implemented a two-stage framework using RCNN for optic disc detection followed by a deep network for glaucoma classification. The model achieved an accuracy of 95.45%, specificity of 93.45%, and sensitivity of 96.43%, demonstrating its effectiveness in automated glaucoma diagnosis.

[13] Test 10: Burlina P et al. (2016)

An SVM classifier was used directly on extracted features from fundus images. The model achieved an accuracy of 95.00%, specificity of 95.60%, and sensitivity of 96.40%, showing strong performance without the need for deep learning.

[19] Test 11: Tan J H et al. (2018)

A deep CNN was employed, achieving an accuracy of 93.45%, specificity of 96.43%, and sensitivity of 79.04%. The model demonstrated high specificity but lower sensitivity, indicating a need for improvement in detecting positive cases.

[18] Test 12: Chai et al. (2018)

This test used an RCNN architecture for glaucoma detection, achieving an accuracy of 91.51%, specificity of 90.90%, and sensitivity of 92.33%. The model performed well but slightly lagged behind deeper architectures.

[16] Test 13: Li et al. (2018)

The Inception-V3 architecture was employed, trained on a large dataset of approximately 40,000 fundus images. The model achieved an AUC of 99%, showcasing state-of-the-art performance in glaucoma detection.

[17] Test 14: Christopher et al. (2018)

A deep learning model was trained on 14,000 fundus images, achieving an AUC of 91%. The model demonstrated strong performance but was outperformed by more advanced architectures like Inception-V3.

Conclusion:

- The proposed models (Tests 1 and 2) achieved 100% specificity, meaning no false positives for normal cases, which is critical for medical diagnostics.
- The Inception-V3 architecture (Test 13) achieved the highest AUC (99%), but it required a significantly larger dataset.
- The SVM-based models (Tests 6 and 10) consistently showed strong performance, particularly in terms of specificity and sensitivity due to the usage of Multi-branch Neural Network in test 6 and SVM Classifier in test 10.

- With 30% ONH crop, the performance remains high ($AUC = 0.84$), indicating that partial removal of the ONH does not significantly degrade performance. Even with 60

2.5 State of the Art Methodology

Key Contributions in Fundus Glaucoma Detection

Architectural Innovations

- Implementation of **end-to-end deep learning frameworks**:
 - Development of a **13-layer CNN** for glaucoma detection using fundus images, achieving high accuracy (93.86% with SoftMax, 95.61% with SVM).
 - Use of **ResNet-50** for both glaucoma detection and vertical cup-to-disc ratio (VCDR) estimation, demonstrating robust performance (AUC: 0.94 for glaucoma detection, R^2 : 77% for VCDR estimation).
 - Exploration of **transfer learning** with pre-trained models (e.g., AlexNet) for glaucoma classification, achieving competitive results (88.2% accuracy).
- Introduction of **explainable AI** for glaucoma detection:
 - Use of **saliency maps** and **occlusion techniques** to visualize regions of interest in fundus images, highlighting the importance of peripapillary areas outside the optic nerve head (ONH).
 - Application of **Remove And Retrain (ROAR)** methodology to objectively evaluate the importance of specific image regions for glaucoma detection and VCDR estimation.

Data Enhancement Techniques

- Application of **data augmentation techniques**:
 - Expansion of the dataset from 1,113 images to 12,012 images using techniques like brightness adjustment, gamma correction, and flipping.
 - Use of **Synthetic Minority Oversampling Technique (SMOTE)** to address class imbalance, ensuring equal representation of normal and glaucomatous images.
- Standardization of fundus images:
 - Resizing and normalization of fundus images to a common resolution (512x512 pixels) for consistency and computational efficiency.
 - Application of **local contrast enhancement** and **circular cropping** to focus on the optic nerve head (ONH) and peripapillary regions.

Experimental Framework

- Comprehensive evaluation across multiple datasets:
 - Training and testing on **multiple datasets** (e.g., HRF, Origa, Drishti, LF Hospital, UZL, REFUGE) to ensure model generalizability.
 - Validation of models on **external datasets** (e.g., REFUGE) to demonstrate cross-dataset performance (AUC: 0.87 on REFUGE).

- Performance metrics and explainability:
 - Evaluation of models using **nine performance metrics**, including accuracy, sensitivity, specificity, precision, AUC, and R^2 for VCDR estimation.
 - Demonstration of **state-of-the-art performance** in glaucoma detection (AUC: 0.94 on UZL) and VCDR estimation (R^2 : 77% on UZL).
 - Analysis of model performance under **extreme cropping conditions** (e.g., 60% ONH crop), showing significant performance even without access to the ONH (AUC: 0.87 for glaucoma detection, R^2 : 37% for VCDR estimation).
- Saliency and occlusion analysis:
 - Use of **averaged saliency maps** to identify recurrent patterns in fundus images, highlighting the importance of inferotemporal and superotemporal regions for glaucoma detection.
 - Application of **occlusion testing** to quantify the impact of removing specific regions (e.g., ONH) on model performance, providing insights into the decision-making process of deep learning models.

2.5.1 Deep Learning Approaches

- **Inception-V3 Architecture:**
 - [16] Li et al. (2018) employed the Inception-V3 architecture, a deep CNN pretrained on large datasets.
 - Achieved an **AUC of 99%** on a dataset of approximately **40,000 fundus images**.
 - Demonstrated the power of **transfer learning** and **large-scale datasets** for glaucoma detection.
- **Custom CNNs:**
 - [3] Raghavendra et al. (2018) developed a **4-layer CNN** with **batch normalization**.
 - Achieved **98.13% accuracy** on a private dataset.
 - Highlighted the effectiveness of **custom architectures** tailored for glaucoma detection.
- **Two-Stage Frameworks:**
 - [12] Bajwa et al. (2019) proposed a **two-stage framework** using **RCNN for optic disc detection**, followed by a deep network for classification.
 - Achieved **95.45% accuracy**, **93.45% specificity**, and **96.43% sensitivity**.
 - Showcased the benefits of **combining localization and classification**.
- **Pretrained Models:**
 - [4] Baida Al Bander et al. (2018) utilized a **pretrained AlexNet model** with an **SVM classifier**.
 - Achieved **88.2% accuracy**, **90.8% specificity**, and **85% sensitivity**.
 - Demonstrated the potential of **combining pretrained models with traditional classifiers**.

2.5.2 Proposed Model Advantages

- The proposed 13-layer CNN with SoftMax and SVM classifiers addresses several of these challenges:
 - High Specificity and Precision: Both versions of the proposed model achieved 100 percent specificity and 100 percent precision, ensuring no false positives for normal cases. This is critical for clinical applications where false positives can lead to unnecessary treatments.
 - Competitive Performance: The CNN-SVM model achieved an accuracy of 95.61 percent, sensitivity of 89.58 percent, and AUC of 94.79 percent, outperforming many state-of-the-art models in terms of accuracy and sensitivity.

2.6 Projects In the Industry

- **Google DeepMind and Moorfields Eye Hospital:** DeepMind, a subsidiary of Google, has partnered with Moorfields Eye Hospital NHS Foundation Trust to develop artificial intelligence (AI) models capable of detecting glaucoma and other eye diseases. Their research focuses on analyzing retinal scans, particularly Optical Coherence Tomography (OCT) images, to identify early signs of glaucoma. Moorfields Eye Hospital is a leader in vision health research and is actively working on implementing AI advancements into practical healthcare applications. MOORFIELDS.NHS.UK
- **DeepMind AI Model Performance:** DeepMind's AI has demonstrated remarkable accuracy in detecting over 50 eye diseases, including glaucoma, with a reported accuracy of 94%. The model was trained on 14,844 retinal scans from approximately 7,500 patients and showed comparable performance to expert ophthalmologists in recommending proper referrals. This AI-based approach has the potential to revolutionize early glaucoma diagnosis, aiding clinicians in assessment and triage. Future steps include clinical trials to evaluate the model's real-world impact and integration into healthcare workflows. MEDTECHDIVE.COM

Chapter 3

Methodology

3.1 Overview

This chapter outlines the methodological framework for developing a deep learning-based glaucoma detection system. The study employs a comparative analysis of convolutional neural network (CNN) architectures to evaluate their effectiveness in classifying fundus images. The pipeline incorporates standardized medical image preprocessing techniques followed by data augmentation strategies to enhance model generalization. Transfer learning principles are applied to leverage pre-trained vision models, which are fine-tuned for the binary classification task. The training process utilizes established optimization techniques while incorporating mechanisms to prevent overfitting. Model performance is quantitatively assessed using standard evaluation metrics, with particular attention to the confidence estimates generated for clinical interpretability. The comparative approach allows for systematic examination of different architectural paradigms in addressing this medical imaging challenge.

3.2 ResNet50

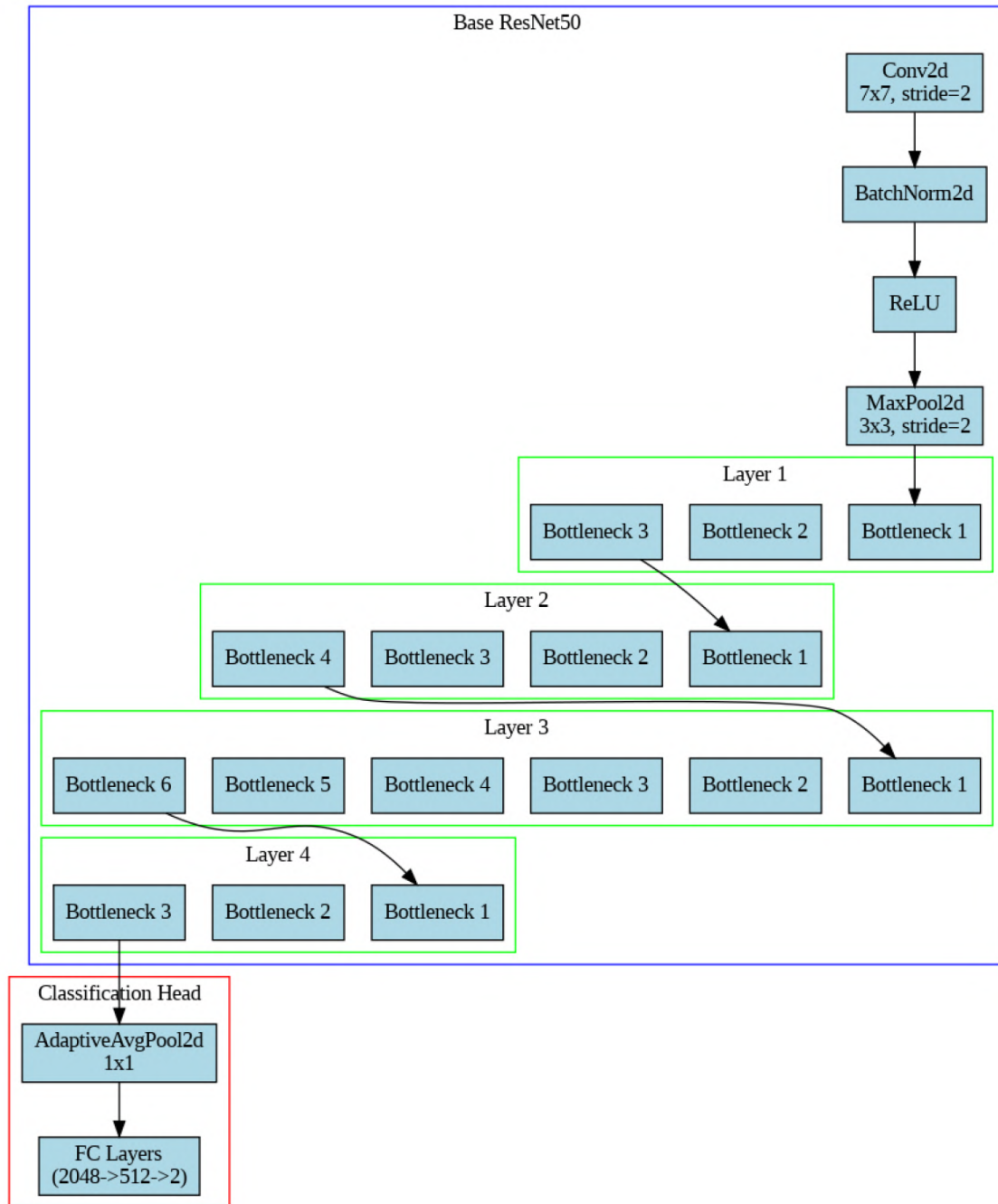


Figure 3.1: ResNet50 Archeticture

Parameter Freezing:

All parameters in the base ResNet50 model (pre-trained on ImageNet) are initially frozen (`requires_grad = False`) to preserve the learned low- and mid-level feature extractors that detect generic visual patterns (edges, textures, shapes). This prevents destructive updates to these generally applicable features during early training phases.

Selective Layer Unfreezing:

Only the final residual block (`layer4`) and the fully-connected head are unfrozen for fine-tuning. This strategic unfreezing allows the model to:

- Adapt high-level feature representations specifically for fundus image characteristics
- Modify spatial hierarchies relevant to pathological structures (optic disc, neuroretinal rim)
- Maintain stability in early layers that capture domain-agnostic features

Theoretical Justification:

This design follows the established transfer learning paradigm where:

- Early convolutional blocks serve as generic feature extractors (Yosinski et al., 2014)
- Later layers require adaptation to domain-specific features (medical imaging)
- Custom heads optimize for task-specific decision boundaries (binary classification)

3.2.1 Convolutional Layer (`conv1`)

The first layer of ResNet is a two-dimensional convolutional layer (`Conv2d`) configured with the following parameters:

- **Input Channels:** 3, corresponding to the three RGB color channels.
- **Output Channels:** 64, producing 64 distinct feature maps.
- **Kernel Size:** 7×7 , selected to capture broad spatial patterns across relatively large receptive fields.
- **Stride:** 2, which reduces the spatial resolution by a factor of two, mitigating computational demands.
- **Padding:** 3, applied symmetrically to preserve spatial dimensions during convolution as per the formula:

$$\text{Output_Size} = \left\lfloor \frac{\text{Input_Size} + 2 \times \text{Padding} - \text{Kernel_Size}}{\text{Stride}} \right\rfloor + 1$$

For an input image size of 224×224 , the output dimensions after this layer are $112 \times 112 \times 64$.

- **Bias Term:** Disabled (`bias=False`), as the subsequent Batch Normalization layer introduces trainable shift parameters, rendering an explicit bias term unnecessary.

Theoretical Note: A larger kernel size at this early stage allows the model to aggregate spatial information over a wider area, which is beneficial for detecting large-scale structures such as edges, blobs, and color gradients.

3.2.2 Batch Normalization Layer (bn1)

Following the convolution, a Batch Normalization layer (`BatchNorm2d`) is applied with the following settings:

- **Number of Features:** 64, matching the output channels of the preceding convolution.
- **Epsilon** ($\varepsilon = 10^{-5}$): A small constant added to the variance to prevent division by zero during normalization.
- **Momentum** (momentum = 0.1): Governs the update rate of running mean and variance estimates, balancing stability and adaptability.
- **Affine Transformation:** Enabled (`affine=True`), allowing learnable scaling (γ) and shifting (β) parameters.
- **Tracking Running Statistics:** Enabled (`track_running_stats=True`), so that global estimates are available for inference.

Theoretical Note: Batch Normalization standardizes the activations within each mini-batch, improving training stability and accelerating convergence by mitigating the problem of internal covariate shift. The epsilon parameter ensures numerical stability by avoiding divisions by very small variances, a technique similarly utilized in optimizers like Adam. Momentum controls the responsiveness of the running estimates: lower values adapt more quickly to changes, while higher values smooth the updates over time.

3.2.3 Activation Function (ReLU)

A Rectified Linear Unit (ReLU) activation is applied element-wise to the normalized output:

$$\text{ReLU}(x) = \max(0, x)$$

The implementation typically uses `inplace=True` to optimize memory usage by modifying the input tensor directly rather than allocating additional memory.

Theoretical Note: ReLU introduces non-linearity into the network, enabling it to approximate complex functions beyond linear mappings. Furthermore, it alleviates the vanishing gradient problem associated with sigmoid and tanh activations, thus facilitating efficient back-propagation and faster learning.

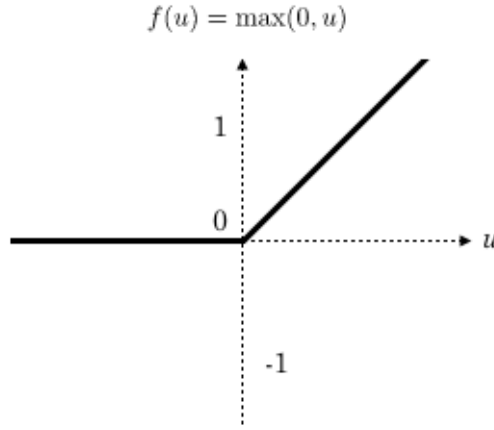


Figure 3.2: Relu Activation Function Figure

3.2.4 Max Pooling Layer (maxpool)

Finally, a max-pooling operation (`MaxPool2d`) further reduces the spatial dimensions of the feature maps:

- **Kernel Size:** 3×3
- **Stride:** 2, reducing the spatial resolution by half.
- **Padding:** 1, ensuring appropriate spatial dimension reduction.

For the input size of $112 \times 112 \times 64$, the output after max pooling becomes $56 \times 56 \times 64$, computed using:

$$\text{Output_Size} = \left\lfloor \frac{112 + 2 \times 1 - 3}{2} \right\rfloor + 1 = 56$$

Theoretical Note: Max pooling serves two main purposes: it reduces computational complexity by down-sampling the feature maps, and it introduces a degree of translation invariance by selecting the most prominent features within each local region.

3.2.5 Bottleneck Block Architecture:

1. The 3-Layer Sandwich Structure Each bottleneck block contains three convolutional layers in this specific order:

- **1×1 Convolution (Channel Reduction):**
 - Reduces input channels (e.g., $256 \rightarrow 64$)
 - **Purpose:** Compress features cheaply before expensive 3×3 operations
 - **Example:** `Conv2d(256, 64, kernel_size=(1,1))`
- **3×3 Convolution (Spatial Processing):**
 - Extracts spatial features (edges, textures, patterns)
 - Maintains or reduces resolution (using `stride=2` in the first blocks)
 - **Example:** `Conv2d(64, 64, kernel_size=(3,3), padding=1)`

Max Pooling Layer (maxpool)



Figure 3.3: MaxPooling Figure

- **1×1 Convolution (Channel Expansion):**

- Expands channels back (e.g., $64 \rightarrow 256$)
- **Purpose:** Restore channel depth for residual connection
- **Example:** `Conv2d(64, 256, kernel_size=(1,1))`

2. Expansion Ratio ($4\times$) The bottleneck ratio defines how much the channels are compressed and expanded.

- **Example in Layer1:**

Input: 256 channels $\rightarrow$ Compressed to 64 (1×1 conv) $\rightarrow$ Expanded back to 256 (1×1 conv)

- **Ratio:** $256 \rightarrow 64 \rightarrow 256$ ($4\times$ compression/expansion)

Why?

- **Saves computation:** The 3×3 convolution operates on fewer channels (64 vs. 256)
- **Preserves representational power:** By expanding back to the original depth

3. Downsampling (Stride=2 in First Blocks) The first bottleneck block in each layer (Layer1–4) uses `stride=2` in its 3×3 convolution to halve spatial dimensions:

Input: 56×56 (Layer1) $\longrightarrow$ Output: 28×28 (Layer2)

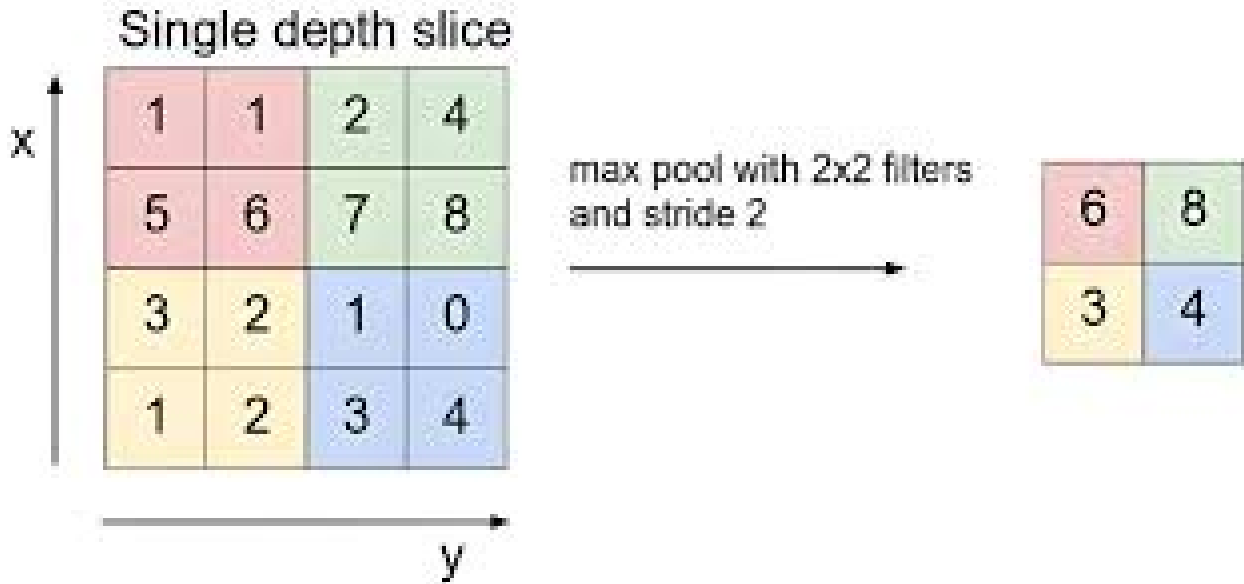


Figure 3.4: Stride 2 Visualization Example

3.2.6 Enhanced Classification Head

1. Linear (2048→512) + BatchNorm + ReLU Why Added:

- **Dimensionality Reduction:** The $2048 \rightarrow 512$ compression reduces the high-dimensional features to a more manageable size before final classification.
- **BatchNorm:** Stabilizes training by normalizing activations, allowing higher learning rates.
- **ReLU Activation:** Introduces non-linearity to capture complex patterns in the reduced space.

Benefits:

- Prevents overfitting by reducing the parameter count before the final layer.
- Makes the model less sensitive to initialization and learning rate choices.
- Helps gradients flow better during backpropagation.

2. Dropout ($p = 0.5$) Why Added:

- Regularization technique that randomly zeros 50% of activations during training.

Benefits:

- Prevents co-adaptation of features (forces the network to use all relevant features).
- Significantly reduces overfitting, especially important in medical imaging where datasets are often small.
- Acts as model averaging during training.

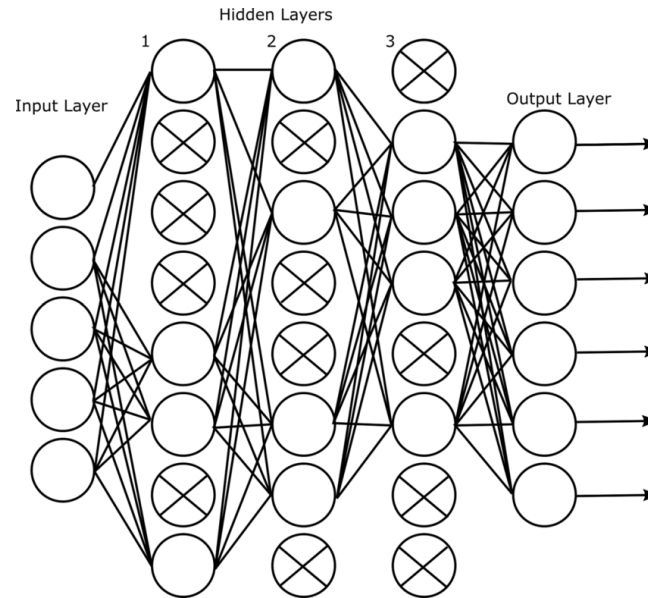


Figure 3.5: DropOut Figure

3. Final Linear (512→2) Why This Design:

- The gradual reduction ($2048 \rightarrow 512 \rightarrow 2$) is gentler than the standard ResNet's direct $2048 \rightarrow 1000$ (for ImageNet).
- Preserves more task-specific information before final classification.

Benefits for Glaucoma Detection:

- Maintains more discriminative features for binary classification.
- The 512D bottleneck layer acts as an "information filter" for the final decision.
- Reduces the risk of losing subtle but important features in fundus images.

Comparison to Standard ResNet50 Head:

Feature	Standard ResNet50	Enhanced Head
Pooling Output	2048D vector	2048D vector
Intermediate Layer	None	512D with BN + ReLU
Regularization	None	Dropout (50%)
Final Layer	Direct to 1000 classes	Gradual reduction to 2 classes
Param Count	$\sim 2.05\text{M}$ (2048×1000)	$\sim 1.05\text{M}$ total (more efficient)

Table 3.1: Comparison between Standard ResNet50 and the Enhanced Classification Head

3.3 MobileNetV3

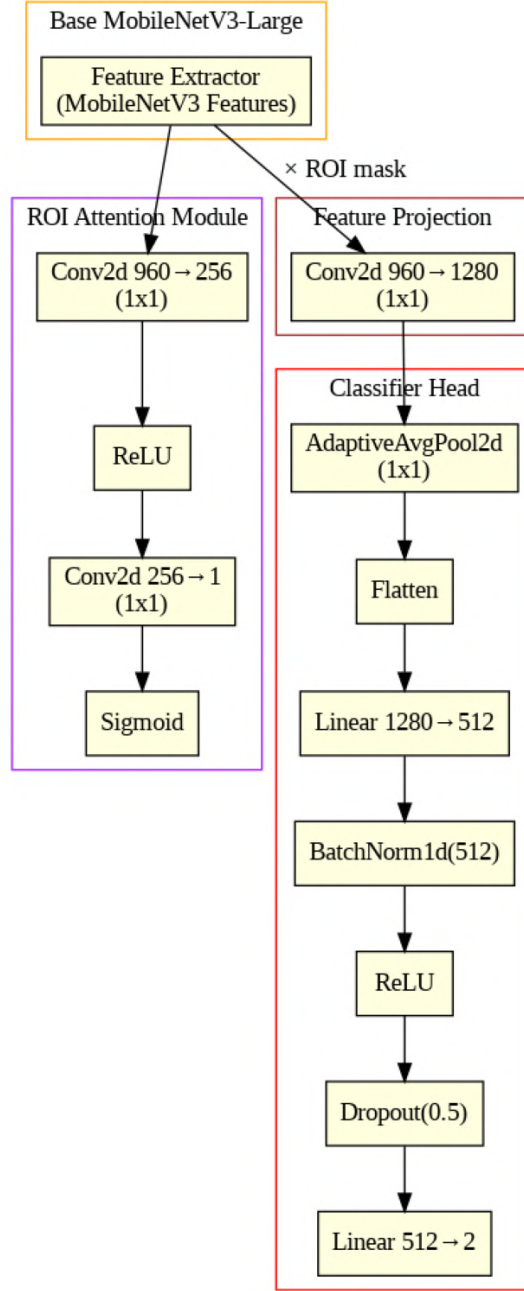


Figure 3.6: MobileNetV3 Archeticture

3.3.1 Initial Convolution

The initial layer of MobileNetV3 consists of a convolutional block designed to efficiently extract low-level features from the input image while reducing its spatial resolution. Specifically, a 3×3 convolution with a stride of 2 and padding of 1 is applied to map the three input channels (RGB) to 16 output channels without a bias term, as subsequent batch normalization will eliminate the need for it. Batch Normalization is then employed to stabilize the training process by normalizing the feature distributions, accelerating convergence and improving generalization. Following normalization, the Hardswish activation function introduces non-linearity

in a computationally efficient manner, approximating the Swish function with piecewise-linear operations to maintain high model performance while being optimized for mobile and edge devices. Together, these operations form a foundational feature extraction block that balances accuracy and efficiency, which is critical for the lightweight design philosophy of MobileNetV3.

3.3.2 Inverted Residual Block Analysis:

1. Use of Conv2d

In MobileNetV3, Conv2d layers serve as fundamental operations for extracting and transforming spatial features. The network heavily relies on two types of convolutions:

- **Depthwise convolution (3×3 , 5×5):** Applies a single convolutional filter per input channel, dramatically reducing computation compared to standard convolution.
- **Pointwise convolution (1×1):** Combines features across channels, allowing the network to mix spatial and channel information effectively.

This separation between spatial and channel processing enables MobileNetV3 to maintain a lightweight structure while preserving sufficient expressive power for accurate prediction. Conv2d thus provides the basic mechanism for feature extraction and transformation with an optimized computational footprint suitable for mobile and embedded devices.

2. Use of BatchNorm2d

BatchNorm2d is applied after each convolutional layer to normalize the activations across a mini-batch. This normalization mitigates the problem of internal covariate shift, allowing the network to train faster and more reliably. Additionally, BatchNorm acts as a regularizer, reducing overfitting and making the model more stable during optimization.

In a lightweight model like MobileNetV3, where parameter efficiency is crucial, BatchNorm2d also helps maintain activation stability across layers without adding significant computational overhead, enabling the use of higher learning rates and more aggressive training strategies.

3. Use of Activation Functions: ReLU and Hardswish

ReLU ReLU (Rectified Linear Unit) is employed primarily in the early stages of MobileNetV3. Its mathematical simplicity—setting all negative values to zero—provides:

- Fast computation
- Sparse activations
- Prevention of vanishing gradient problems

ReLU is highly efficient, making it a logical choice for the shallow parts of the network where computational cost is more critical than expressive capacity.

Hardswish Hardswish is a computationally cheaper approximation of the swish activation function, defined as:

$$\text{Hardswish}(x) = x \cdot \frac{\text{ReLU6}(x + 3)}{6}$$

Hardswish provides a smooth non-linearity, helping to maintain richer information through the network without introducing heavy computation. It is particularly used in deeper parts of the network where the expressiveness of activations becomes more important for achieving high accuracy.

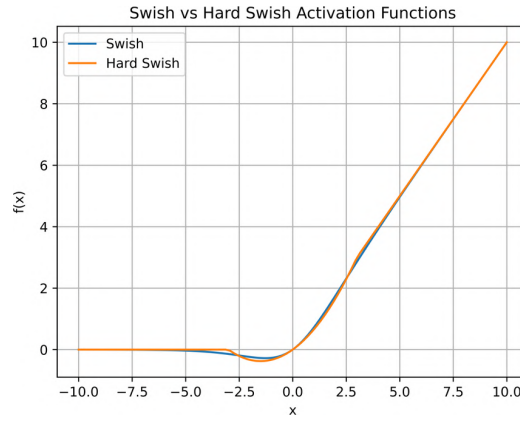


Figure 3.7: Hardswish Activation Function Figure

4. Use of SE (Squeeze-and-Excitation) Modules

The Squeeze-and-Excitation (SE) modules are integrated selectively in MobileNetV3 to improve the representational capacity of the network by explicitly modeling channel-wise interdependencies.

The SE block consists of three steps:

- **Squeeze:** Global average pooling condenses each feature map to a single value, summarizing global spatial information.
- **Excitation:** A small fully connected network (often a bottleneck with reduction ratio like 4 or 8) learns non-linear interactions between channels and outputs channel-wise weights.
- **Scale:** The original feature maps are reweighted (scaled) by these learned channel weights.

By recalibrating feature responses, the SE module allows the network to focus more on important channels and suppress less useful ones, improving overall feature quality without a significant increase in computational cost.

In MobileNetV3, SE blocks are used selectively based on the design trade-offs between efficiency and accuracy: they are mainly inserted after convolutions in blocks where additional channel-wise attention provides a noticeable accuracy boost for a small computational cost.

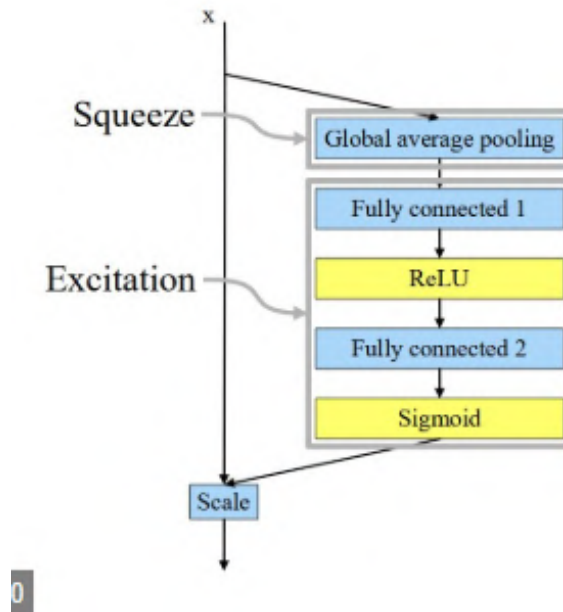


Figure 3.8: Squeeze Excitation Modules Figure

5. Conv2dNormActivation Block

Structure:

```

Conv2d(160, 960, kernel_size=1, stride=1, bias=False)
BatchNorm2d(960)
Hardswish()

```

Explanation:

- **Conv2d (1×1 convolution):** Increases the number of channels from 160 to 960 without changing spatial dimensions.
Purpose: Performs a channel expansion to allow richer representations and more complex features, while keeping computational cost low compared to larger kernels.
- **BatchNorm2d:** Normalizes the output for stability and faster training.
Purpose: Controls activation distribution across the batch, speeding up convergence and acting as slight regularization.
- **Hardswish:** Provides a smooth, lightweight non-linearity.
Purpose: Enables richer features without heavy computation, essential at deeper stages where expressive power matters.

Summary: This block prepares strong, rich feature maps before the pooling and classification stages by increasing the channel capacity with stable, non-linear transformations.

5. Adaptive Average Pooling Layer

Structure:

```

AdaptiveAvgPool2d(output_size=1)

```


Explanation:

- **AdaptiveAvgPool2d:** Reduces the spatial dimension of feature maps to 1×1 regardless of input size.

Purpose: Compresses spatial information globally, leaving only channel descriptors essential for final prediction layers.

Summary: Turns large feature maps into compact, globally-aware vectors ready for classification.

3.4 EfficientNetV2-S

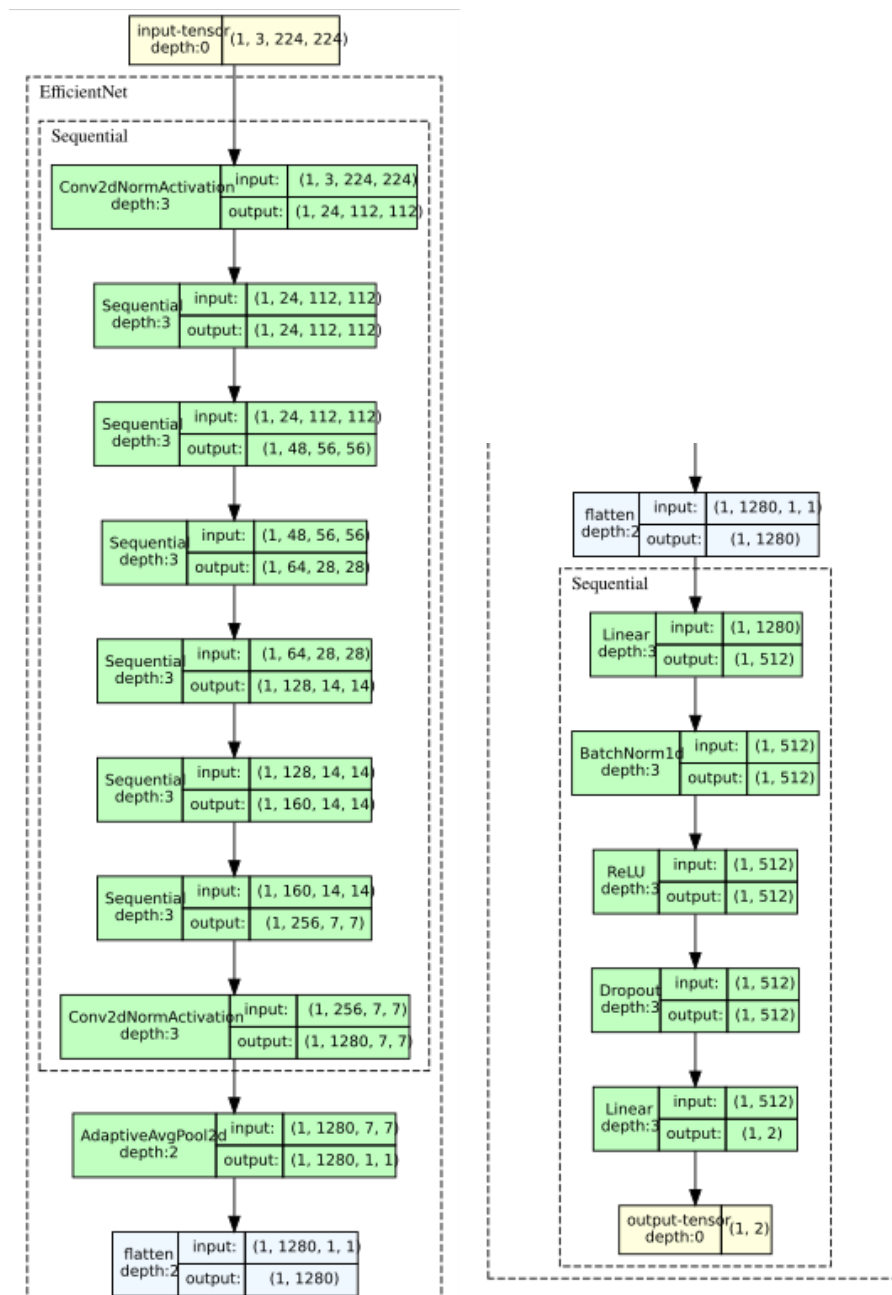


Figure 3.9: Split view of EfficientNetV2-S Architecture

3.4.1 Initial Convolution Layer (Stage 0)

Description

The initial convolution layer is the entry point of the model, transforming input images (3 channels: RGB) into a higher-dimensional feature map with 24 channels. It employs a 3x3 convolutional kernel with a stride of 2, reducing spatial dimensions by half, followed by batch normalization and SiLU (Swish) activation.

Functionality

- The convolution extracts low-level features (e.g., edges, textures) by applying learnable filters.
- Stride-2 reduces spatial resolution, decreasing computational load early in the network.
- Batch normalization stabilizes training by normalizing activations, and SiLU introduces non-linearity to enhance feature expressiveness.

Theoretical Importance

- Serves as the foundation for feature extraction, capturing fundamental patterns critical for subsequent layers.
- The stride-2 convolution balances computational efficiency with information retention, a key principle in efficient CNN design.
- SiLU activation, defined as

$$f(x) = x \cdot \sigma(x),$$

improves gradient flow compared to ReLU, enhancing convergence and model performance.

3.4.2 FusedMBConv Blocks (Stages 1–3)

Description

The Fused Mobile Inverted Bottleneck Convolution (FusedMBConv) blocks are used in stages 1–3, comprising lightweight convolutional modules. Each block includes a 3x3 convolution (sometimes followed by a 1x1 convolution for channel reduction), batch normalization, SiLU activation, and StochasticDepth for regularization. Stages 1–3 progressively increase channel sizes (24→48→64).

Functionality

- The 3x3 convolution performs spatial feature extraction, while the optional 1x1 convolution reduces channel dimensions for efficiency.
- Batch normalization and SiLU ensure stable and expressive feature representations.
- StochasticDepth randomly drops residual connections during training, acting as a regularizer to prevent overfitting.

Theoretical Importance

- FusedMBConv blocks optimize computational efficiency by combining expansion and projection convolutions into a single operation, reducing memory access costs compared to traditional MBConv blocks.
- They extract mid-level features (e.g., shapes, patterns), critical for building hierarchical representations in CNNs.
- StochasticDepth enhances generalization by introducing randomness, akin to dropout, particularly effective in deep networks.

3.4.3 MBConv Blocks (Stages 4–6)

Description

Mobile Inverted Bottleneck Convolution (MBConv) blocks form the core of stages 4–6, employing a more complex structure with depthwise separable convolutions. Each block consists of a 1x1 expansion convolution, a 3x3 depthwise convolution, a Squeeze-and-Excitation (SE) module, a 1x1 projection convolution, batch normalization, SiLU activation, and StochasticDepth. Channel sizes increase from 64 to 256 across these stages.

Functionality

- The 1x1 expansion convolution increases channel dimensions, enhancing feature richness.
- The 3x3 depthwise convolution applies spatial filtering independently per channel, reducing parameters compared to standard convolutions.
- The SE module recalibrates channel importance via global average pooling and two fully connected layers, improving feature discriminability.
- The 1x1 projection convolution reduces channel dimensions, and StochasticDepth regularizes training.

Theoretical Importance

- MBConv blocks leverage depthwise separable convolutions to achieve a balance between computational efficiency and expressive power.
- The SE module introduces attention-like mechanisms, allowing the model to focus on relevant features, which is critical for fine-grained discrimination.
- These blocks extract high-level, task-specific features, forming the backbone of the model’s ability to learn complex patterns.

3.4.4 Final Convolution Layer (Stage 7)

Description

The final convolution layer is a 1x1 convolution that projects the 256-channel feature maps from stage 6 to 1280 channels, followed by batch normalization and SiLU activation.

Functionality

- The 1x1 convolution increases channel dimensionality, creating a rich feature representation suitable for classification.
- Batch normalization stabilizes the high-dimensional features, and SiLU enhances non-linearity.

Theoretical Importance

- Acts as a feature aggregation layer, consolidating high-level features before global pooling and classification.
- The high channel count (1280) ensures a comprehensive feature space, capturing diverse patterns necessary for robust classification.
- Maintains model efficiency by using a parameter-light 1x1 convolution, aligning with the design principles of EfficientNet.

3.4.5 Adaptive Average Pooling

Description

The adaptive average pooling layer reduces the spatial dimensions of the feature maps (from stage 7) to 1x1, producing a 1280-dimensional feature vector per image.

Functionality

- Performs global average pooling, averaging each channel across spatial dimensions to create a compact representation.
- Eliminates spatial information, focusing on channel-wise feature strengths.

Theoretical Importance

- Enables the model to handle variable input sizes by producing a fixed-size output, enhancing flexibility.
- Reduces overfitting by summarizing spatial information, acting as a form of regularization.
- Prepares features for the classifier by creating a high-dimensional, spatially invariant representation, a standard practice in modern CNNs.

3.5 Optimization Algorithms

3.5.1 RMSprop (Root Mean Square Propagation)

RMSprop [27] is an adaptive learning rate optimization method designed to address the limitations of vanilla stochastic gradient descent (SGD) by adjusting the learning rate per parameter based on recent gradient magnitudes. It maintains a moving average of squared gradients to normalize the gradient updates.

The update rules for RMSprop are as follows:

1. **Accumulate squared gradients:**

$$E[g^2]_t = \beta E[g^2]_{t-1} + (1 - \beta)g_t^2 \quad (3.1)$$

where g_t is the gradient at time step t , $E[g^2]_t$ is the exponentially decaying average of squared gradients, and β is the decay rate (typically 0.9).

2. **Parameter update:**

$$\theta_{t+1} = \theta_t - \frac{\eta}{\sqrt{E[g^2]_t + \epsilon}} \odot g_t \quad (3.2)$$

where θ_t represents the parameters, η is the learning rate, ϵ is a small constant for numerical stability, and $\odot$ denotes element-wise multiplication.

3.5.2 Momentum

Momentum [28] accelerates gradient descent by incorporating a velocity term that accumulates past gradients, helping to dampen oscillations and speed up convergence.

The update rules for Momentum are:

1. **Velocity update:**

$$v_t = \gamma v_{t-1} + \eta g_t \quad (3.3)$$

where v_t is the velocity at time t , γ is the momentum coefficient (typically 0.9), and η is the learning rate.

2. **Parameter update:**

$$\theta_{t+1} = \theta_t - v_t \quad (3.4)$$

3.5.3 AdamW (Adam with Weight Decay)

AdamW [29] is an improvement over Adam that decouples weight decay from the adaptive gradient updates, leading to more effective regularization.

The update rules for AdamW are:

1. **Compute biased moments:**

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1)g_t \quad (3.5)$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2)g_t^2 \quad (3.6)$$

where m_t and v_t are estimates of the first and second moments of the gradients, and β_1, β_2 are exponential decay rates (typically 0.9 and 0.999).

2. **Bias correction:**

$$\hat{m}_t = \frac{m_t}{1 - \beta_1^t} \quad (3.7)$$

$$\hat{v}_t = \frac{v_t}{1 - \beta_2^t} \quad (3.8)$$

3. **Parameter update (with decoupled weight decay λ):**

$$\theta_{t+1} = \theta_t - \eta \left(\frac{\hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}} + \lambda \theta_t \right) \quad (3.9)$$

Chapter 4

Results & Discussion

4.1 Overview

The Results and Discussion chapter presents a comprehensive evaluation of deep learning models—ResNet50, EfficientNet (with different variants) and MobileNetV2 the binary classification of retinal images into Glaucoma and Normal classes. Each model’s performance was rigorously assessed using multiple metrics, including accuracy, precision, recall, F1 score, Area Under the Curve (AUC), and confusion matrices, alongside Receiver Operating Characteristic (ROC) curves to visualize classification efficacy. The study highlights the comparative strengths and limitations of these architectures in detecting glaucoma, providing insights into their suitability for clinical applications. By analyzing model performance across these metrics, this chapter elucidates the trade-offs between computational efficiency and diagnostic accuracy, contributing to the ongoing discourse on automated glaucoma detection using deep learning.

4.2 Dataset

4.2.1 Image Pre-processing Pipeline for Fundus Images

To enhance the quality and consistency of retinal fundus images prior to model training, a custom preprocessing pipeline was implemented. The key steps in the pipeline are outlined below:

Color Space Transformation (RGB to LAB)

The original fundus images, typically in the RGB color space, were first converted to the LAB color space. This color space separates the lightness information (L channel) from color information (A and B channels). The separation allows for targeted contrast enhancement on the brightness component without distorting the color distribution of the images.

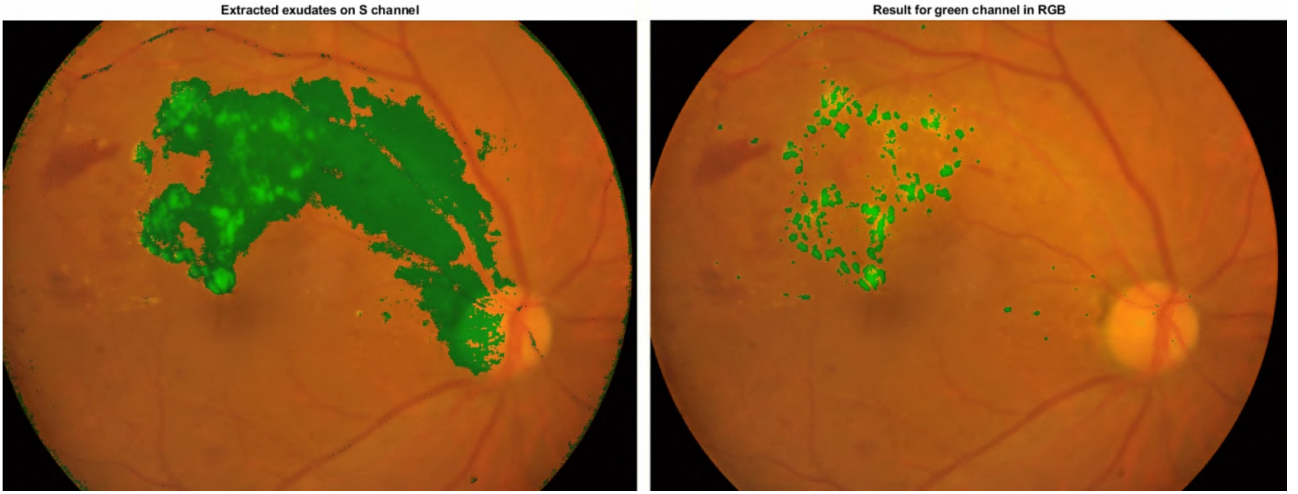


Figure 4.1: Retinal Image RGB to LAB Transformation

Local Contrast Enhancement using CLAHE

Following the conversion, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied to the L channel. CLAHE enhances the local contrast of images by performing histogram equalization on small regions (tiles), rather than globally. This technique improves the visibility of important retinal structures (such as blood vessels and lesions) while avoiding over-amplification of noise, which is critical in medical imaging tasks.

In this setup, the CLAHE parameters were:

- **Clip Limit:** 2.0
- **Tile Grid Size:** (8, 8)

Merging Channels and Reverting to RGB

After enhancing the L channel, the A and B channels were recombined with the modified L channel. The resulting LAB image was then converted back to the RGB color space. This step ensures that while brightness variations were corrected, the chromatic content of the original image remained intact.

Gamma Correction for Data Augmentation

To further increase the variability and robustness of the dataset, a random gamma correction was applied. Gamma values were sampled uniformly between 0.8 and 1.2 for each image. This process simulates slight variations in lighting conditions, thus helping the model to generalize better across different acquisition setups and illumination levels.

The gamma correction operation is defined as:

$$I_{\text{out}}(x, y) = 255 \times \left(\frac{I_{\text{in}}(x, y)}{255} \right)^{\gamma}$$

where γ is the randomly chosen gamma value.

Final Output

The final preprocessed image was returned as a standard RGB image, ready for further transformations (such as normalization or tensor conversion) before being fed into the deep learning model.

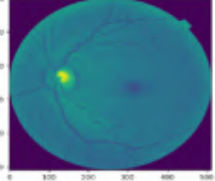
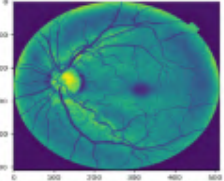
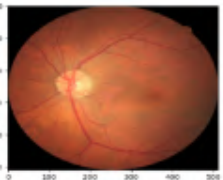
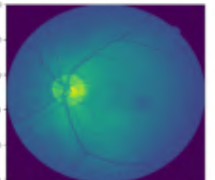
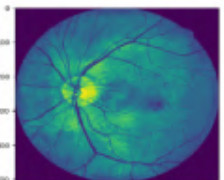
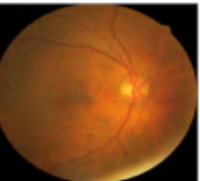
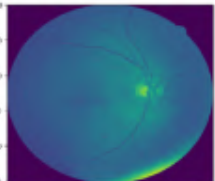
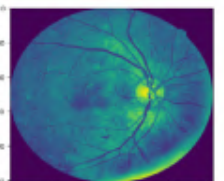
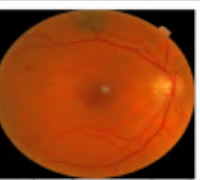
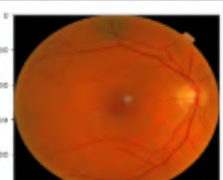
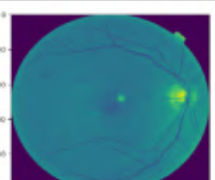
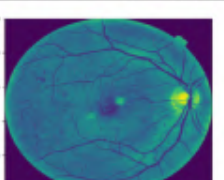
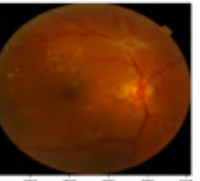
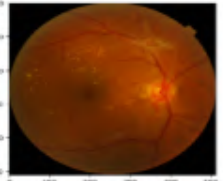
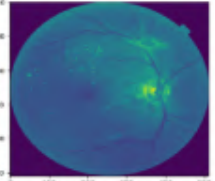
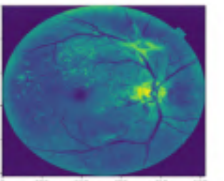
Severity Grading	Input Image	AHE	Gamma Correction	CLAHE
No Category DR				
Mild Category NPDR				
Moderate Category NPDR				
Severe Category NPDR				
PDR				

Figure 4.2: Retinal Image transformations from Resaerch Papers

Summary of Advantages

- **Improved Local Contrast:** Enhances critical fine structures in fundus images.
- **Preserved Color Information:** Maintains diagnostic color cues crucial for pathology detection.
- **Data Augmentation via Gamma Variations:** Increases dataset diversity to improve model generalization.
- **Compatibility:** Outputs standard RGB images, easily integrable into existing deep learning pipelines.

4.2.2 Data Augmentation/Transformation

In order to improve the generalization of the model and increase its robustness to real-world variations in retinal fundus images, several augmentation techniques are applied to the training dataset. These techniques are intended to simulate various conditions under which the images may have been captured, making the model more invariant to different imaging artifacts and conditions. The specific transformations applied are as follows:

Resize

```
transforms.Resize((input_size, input_size))
```

The images are resized to a fixed size of $(input_size, input_size)$ pixels (typically 224x224 pixels). This step ensures that all input images are of consistent size, which is required by deep learning models. In this case, 224x224 is chosen as a standard resolution because:

- **Standardization of Image Size:** Since retinal fundus images can have varying dimensions depending on the device or conditions under which they were captured, resizing all images to a common resolution ensures uniformity. A consistent input size simplifies the model architecture and ensures that the neural network can process the images effectively.
- **Optimization for Pre-trained Models:** 224x224 is a commonly used resolution, particularly for pre-trained models like ResNet50 and others in transfer learning applications. Using a resolution of 224x224 allows the use of pre-trained weights from these models, which have been trained on datasets like ImageNet that use this resolution. By maintaining this standard resolution, we leverage the powerful features learned by these models, improving performance.
- **Computational Efficiency:** Larger image sizes capture more detailed information but also increase computational cost (memory usage and processing time). 224x224 strikes a balance between sufficient detail and computational feasibility, enabling faster training and inference times without excessively compromising information.

Random Rotation

```
transforms.RandomRotation(15)
```

Random rotation is applied to the images, where the image is randomly rotated by an angle in the range of ± 15 degrees. This transformation is useful for simulating variations in the orientation of the retina in fundus images. The retina's features can appear differently depending on how the image was captured (e.g., slight tilts or rotations). By introducing random rotations, the model becomes invariant to the orientation of the input images, allowing it to recognize key retinal features regardless of their angle in the image.

Random Horizontal and Vertical Flip

`transforms.RandomHorizontalFlip()` and `transforms.RandomVerticalFlip()`

These two transformations randomly flip the image horizontally and vertically with a 50% chance. These flips are particularly important in medical imaging, as abnormalities in retinal images may appear in a mirrored form across the horizontal or vertical axis. By applying random flips, the model learns to recognize features regardless of their orientation, improving its ability to generalize across different cases.

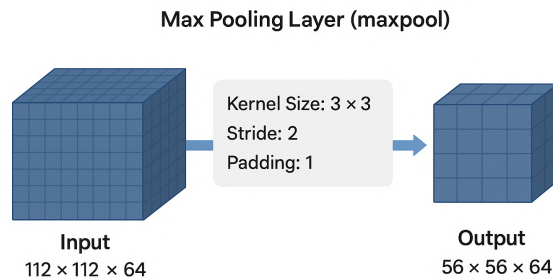


Figure 4.3: MaxPooling Figure

Color Jitter

`transforms.ColorJitter(brightness=0.1, contrast=0.1, saturation=0.1, hue=0.05)`

The `ColorJitter` transformation randomly alters the brightness, contrast, saturation, and hue of the image within a given range:

- **Brightness:** The brightness is randomly adjusted by a factor between 0.9 and 1.1, allowing for subtle variations in lighting.
- **Contrast:** Similarly, the contrast is adjusted by a factor between 0.9 and 1.1, making the image appear slightly more or less contrasted.
- **Saturation:** The color saturation is also adjusted within the range 0.9 to 1.1, simulating slight shifts in color intensity.
- **Hue:** The hue (the actual color of the image) is altered within a range of ± 0.05 , allowing for a minor shift in the color spectrum.

These random changes simulate real-world variations in imaging conditions, such as differences in camera settings, lighting, or environmental factors. These color adjustments are particularly important in medical imaging, where features may appear differently depending on factors like lighting conditions or camera quality.

ToTensor

`transforms.ToTensor()`

This transformation converts the image from a PIL (Python Imaging Library) format to a tensor format. Tensors are the fundamental data structure used in PyTorch, and they are required for feeding the images into deep learning models. This step also normalizes the pixel values from the range of $[0, 255]$ to $[0.0, 1.0]$.

Normalize

```
transforms.Normalize(mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225])
```

Normalization is applied to the image after it has been converted to a tensor. This step adjusts the pixel values by subtracting the mean and dividing by the standard deviation for each of the three RGB channels. The values used for the mean and standard deviation are based on the statistics of the ImageNet dataset, which is commonly used for pre-trained models like ResNet.

- **Mean:** $[0.485, 0.456, 0.406]$
- **Standard Deviation:** $[0.229, 0.224, 0.225]$

Normalization ensures that the input images have a similar distribution to the data used to train the pre-trained models. This helps the model learn more effectively by making the input data distribution match that of the model's expected input.

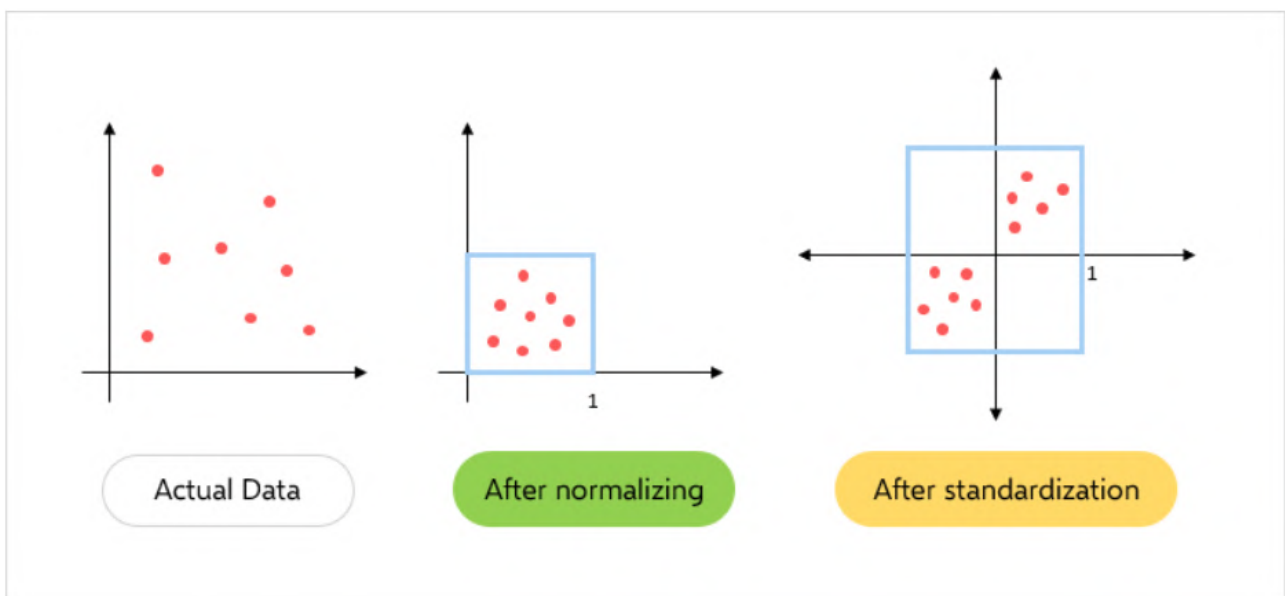


Figure 4.4: Normalization & Standardization Process

Validation Transformation

The validation transformation (`val_transform`) follows a similar structure as the training transformations but with a few key differences:

- **No Random Augmentations:** Unlike the training set, the validation set does not undergo random transformations (such as rotation, flipping, or color jittering) because the goal is to evaluate the model on data that is as close to the real-world distribution as possible.
- **Fixed Resize:** The image is resized to a fixed size (224×224), which ensures consistency across the validation dataset.
- **Normalization:** The images are normalized in the same way as the training images, using the same mean and standard deviation values.

Transformation Visualization:

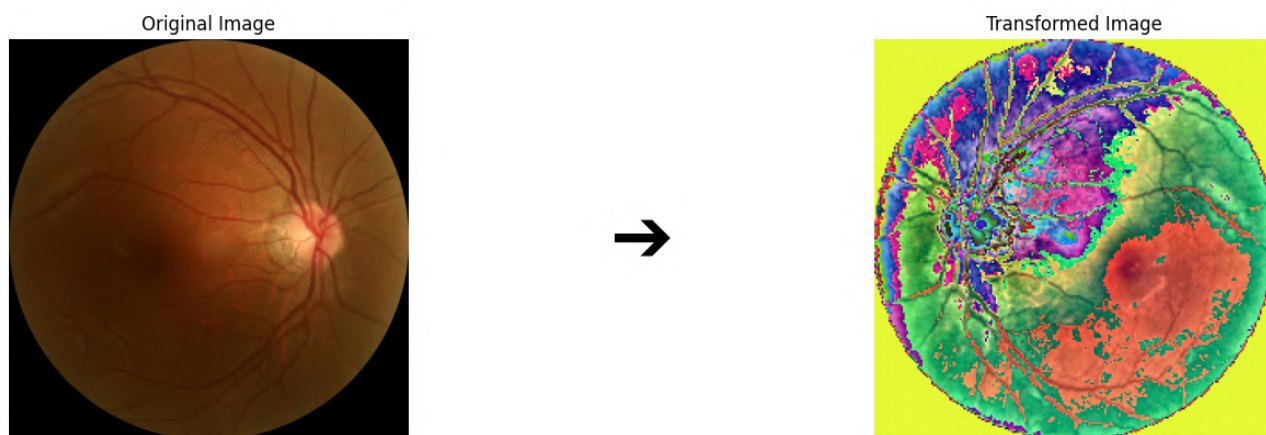


Figure 4.5: Normal Image Transformation

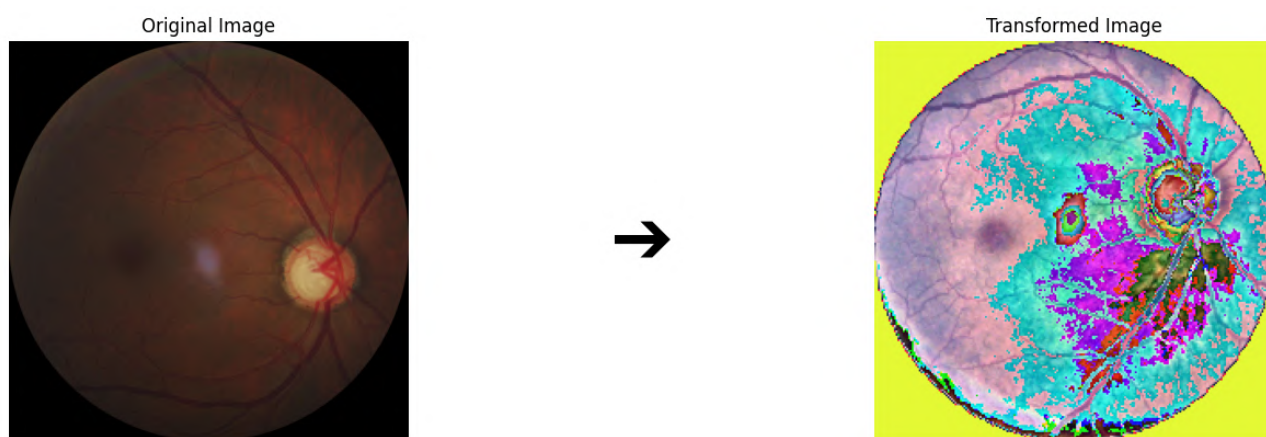


Figure 4.6: Glaucoma Image Transformation

Impact on Model Performance

By incorporating these transformations, the model is trained to recognize retinal features across a variety of image conditions, including different orientations, lighting, and color variations. The primary goal of these augmentations is to reduce overfitting, ensuring that the model learns to generalize well to unseen data.

Through augmentation, the dataset effectively increases in size, exposing the model to a broader spectrum of potential input images, which helps the model learn more robust and invariant features. As a result, the model is better equipped to handle real-world data variability, improving its performance in tasks such as detecting retinal diseases and abnormalities from fundus images.

4.3 Dataset Used

SMDG-19

This study utilizes the Fundus-PyTorch dataset, which contains standardized retinal fundus images from diverse clinical sources. The dataset focuses on binary classification of glaucoma progression, providing essential data for automated detection systems. As previously discussed, the dataset is organized into two clinically significant classes:

- **Normal (Class 0):** Healthy retinal fundus images showing no signs of glaucoma (10,586 images)
- **Glaucoma (Class 1):** Fundus images exhibiting characteristic optic nerve damage and visual field loss associated with glaucoma (6,656 images)

The complete dataset contains 17,242 high-resolution fundus images, offering a robust foundation for training and evaluating deep learning models as examined in the literature review. Table 3.1 presents the detailed distribution of images across dataset splits.

Table 4.1: Distribution of Fundus Images Across Dataset Splits

Split	No Glaucoma	Glaucoma	Total
Train	5,293	3,328	8,621 (50%)
Validation	3,539	2,208	5,747 (33.3%)
Test	1,754	1,120	2,874 (16.66%)
Total	10,586	6,656	17,242

4.3.1 Models Used

- ResNet50
- EfficientNetV2_S
- MobileNetV2

4.4 ResNet50

Layer	Type	Configuration	Output Size	Parameters	Activation
Initial Conv	Conv2d	7x7, stride=2, padding=3, in=3, out=64	112x112x64	9,408	-
BatchNorm2d	64 channels	-	112x112x64	128	-
ReLU	-	inplace=True	112x112x64	-	ReLU
MaxPool	MaxPool2d	3x3, stride=2, padding=1	56x56x64	-	-
Layer1 (x3)	Bottleneck	[1x1,64 → 3x3,64 → 1x1,256] (downsample for first block)	56x56x256	~70,400/block	ReLU
Layer2 (x4)	Bottleneck	[1x1,128 → 3x3,128 → 1x1,512] (stride=2 first block)	28x28x512	~280,000/block	ReLU
Layer3 (x6)	Bottleneck	[1x1,256 → 3x3,256 → 1x1,1024] (stride=2 first block)	14x14x1024	~1,180,000/block	ReLU
Layer4 (x3)	Bottleneck	[1x1,512 → 3x3,512 → 1x1,2048] (stride=2 first block)	7x7x2048	~4,720,000/block	ReLU
Head	AdaptiveAvgPool2d	output_size=(1,1)	1x1x2048	-	-
Linear + BN + ReLU	Linear	2048→512	512	1,049,088	ReLU
Dropout	Dropout	p=0.5	512	-	-
Linear	Linear	512→2 (output)	2	1,026	-

Total Parameters: ~ 25.6 million

4.4.1 Training Configuration and Analysis

The model was trained using a transfer learning approach with a pretrained ResNet-50 backbone. Several techniques were employed to improve generalization and reduce overfitting, ultimately leading to an accuracy of approximately 99%. Below is a summary of the most important aspects of the training pipeline:

Training vs. Validation Accuracy: In some epochs, validation accuracy exceeded training accuracy, likely due to the heavier augmentations applied to the training images, which increase difficulty. This behavior is typical in robust training pipelines and reflects effective regularization.

The combination of aggressive augmentations, transfer learning, regularization techniques, and a carefully designed training pipeline played a pivotal role in achieving a high-performing glaucoma classification model.

Table 4.3: Training Settings and Hyperparameters

Category	Details
Kaggle Device	P100
Epochs	50
Learning Rate	3e-4
Batch Size	32
Optimizer	AdamW
Loss Function	LabelSmoothingCrossEntropy
Scheduler	ReduceLROnPlateau
Regularization	Dropout (0.5)
unfreezing	Last 2 Blocks
LR Milestones	[7, 14, 21, 28, 35]
Weight Decay	1e-5
Early Stopping	Patience: 7, Delta: 0
Model	Enhanced ResNet50
Parameters Used	Total: 24,559,170, Trainable: 16,015,874

4.4.2 Performance Metrics

The performance evaluation in 4.4 of the Enhanced ResNet50 model on the test dataset demonstrated outstanding classification results. The model achieved an overall accuracy of 99% across 2874 samples, indicating highly reliable performance. Class-wise metrics further validate this finding: for the normal (class 0) category, the model attained a precision of 0.98, a recall of 0.99, and an F1-score of 0.99, showcasing its excellent ability to correctly identify healthy cases with almost no false positives. Conversely, for the diseased (class 1) category, the model achieved a precision of 0.99, a recall of 0.97, and an F1-score of 0.98, reflecting its strong capability to detect positive cases with minimal false negatives. Additionally, both the macro-averaged and weighted-averaged metrics recorded values of 0.99, further indicating consistent model performance across classes despite slight class imbalance. These results, complemented by a near-perfect ROC AUC score, substantiate the model's robustness, making it highly suitable for practical deployment in fundus image-based disease detection tasks.

Class	Precision	Recall	F1-Score	Support
0	0.98	0.99	0.99	1754
1	0.99	0.97	0.98	1120
Accuracy			0.99	2874
Macro Avg	0.99	0.98	0.98	2874
Weighted Avg	0.99	0.99	0.99	2874

Table 4.4: Classification report

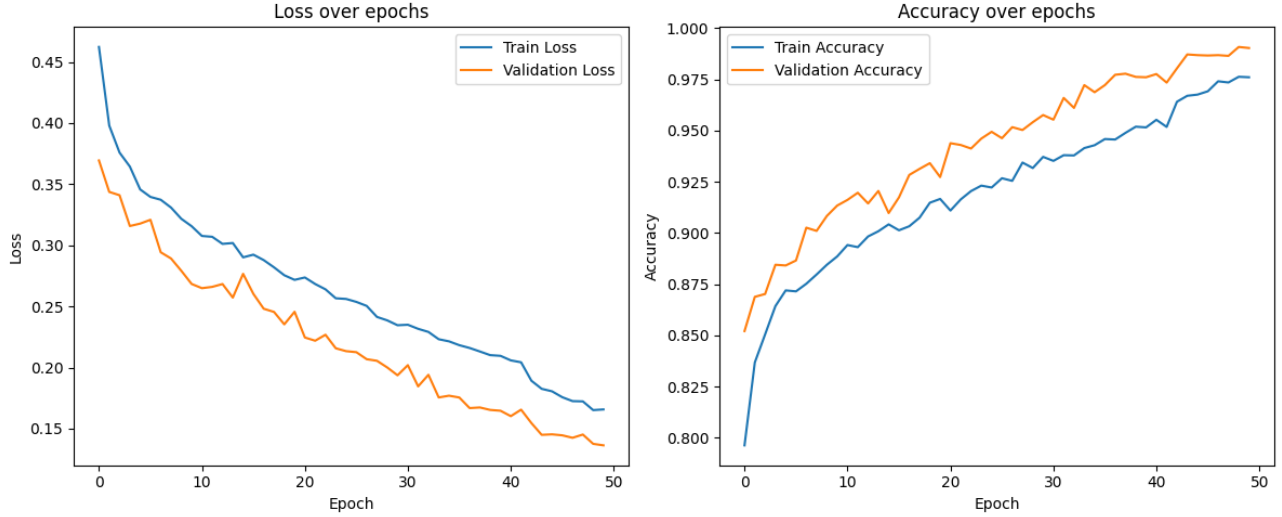


Figure 4.7: Accuracy and loss curves for the ResNet model.

4.4.3 Confusion Matrix Analysis

The confusion matrix in Figure 4.8 provides a detailed breakdown of the classification model's performance. Out of 1754 actual class 0 instances, 1746 were correctly predicted, with only 8 misclassified as class 1, reflecting a very high true negative rate. Similarly, out of 1120 actual class 1 instances, 1093 were correctly identified, and only 27 were incorrectly classified as class 0. These results demonstrate the model's strong predictive capabilities, achieving a high degree of accuracy with minimal misclassifications. The model's ability to correctly distinguish between the two classes indicates that it has learned the underlying patterns effectively, with a slightly better performance on class 0 compared to class 1. The relatively low number of false positives and false negatives suggests a balanced trade-off between precision and recall, making the model robust and reliable for practical applications.

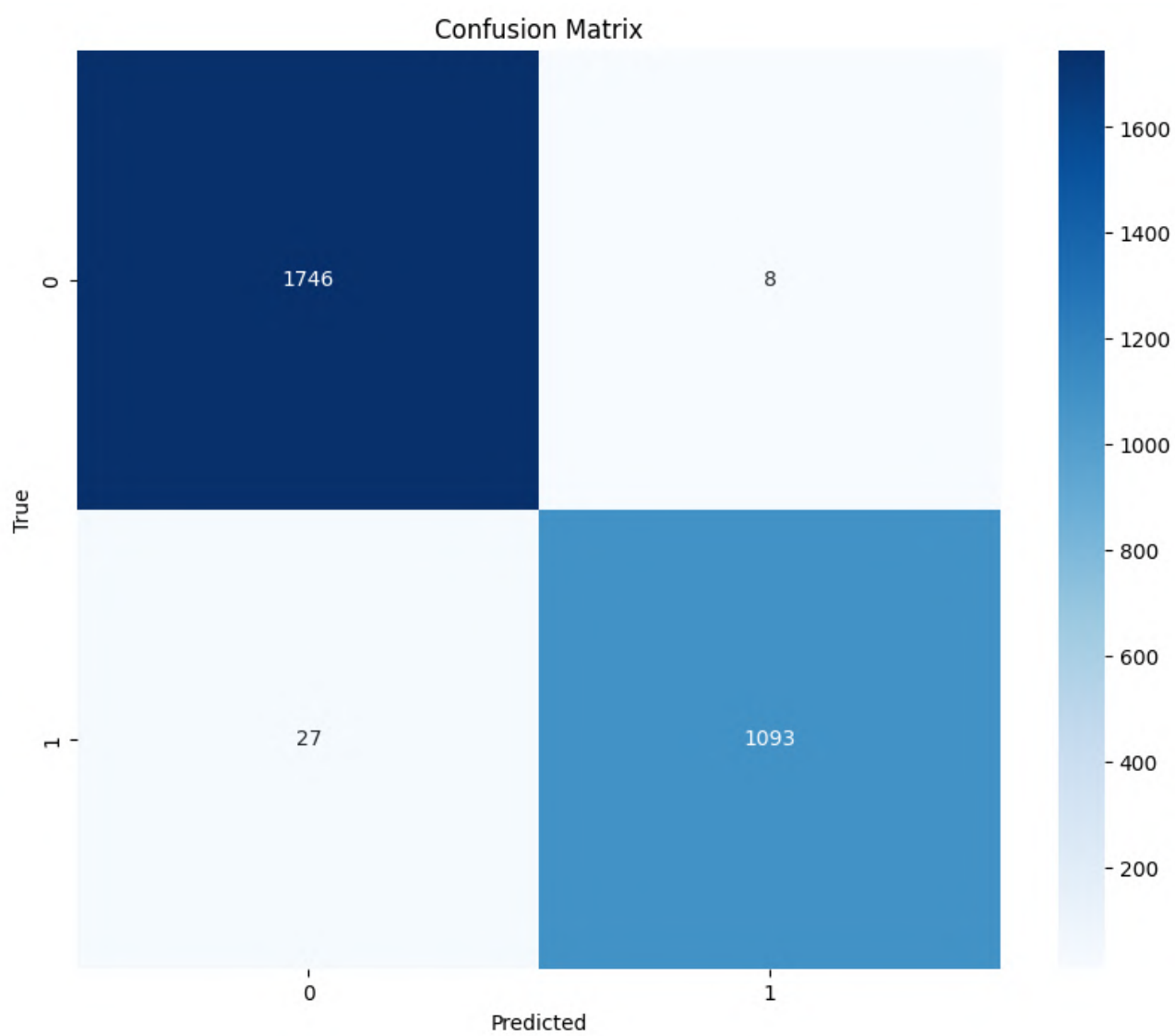


Figure 4.8: Confusion matrix for the ResNet model.

4.4.4 ROC Curve analysis

The ROC curve in Figure 4.4.4 for the Enhanced ResNet50 Fundus Model demonstrates exceptional performance, with an Area Under the Curve (AUC) of 1.00, indicating perfect classification ability. The curve hugs the top-left corner, showing a True Positive Rate (sensitivity) of 1.0 across all False Positive Rate (specificity) thresholds, meaning the model correctly identifies all positive cases without any false positives. This suggests that the model is highly effective in distinguishing between classes in fundus image classification, likely due to the enhancements made to the ResNet50 architecture, which may include improved feature extraction or optimization techniques tailored for medical imaging. Such performance is ideal for clinical applications where both high sensitivity and specificity are critical, ensuring reliable detection of conditions like diabetic retinopathy or other retinal disorders.

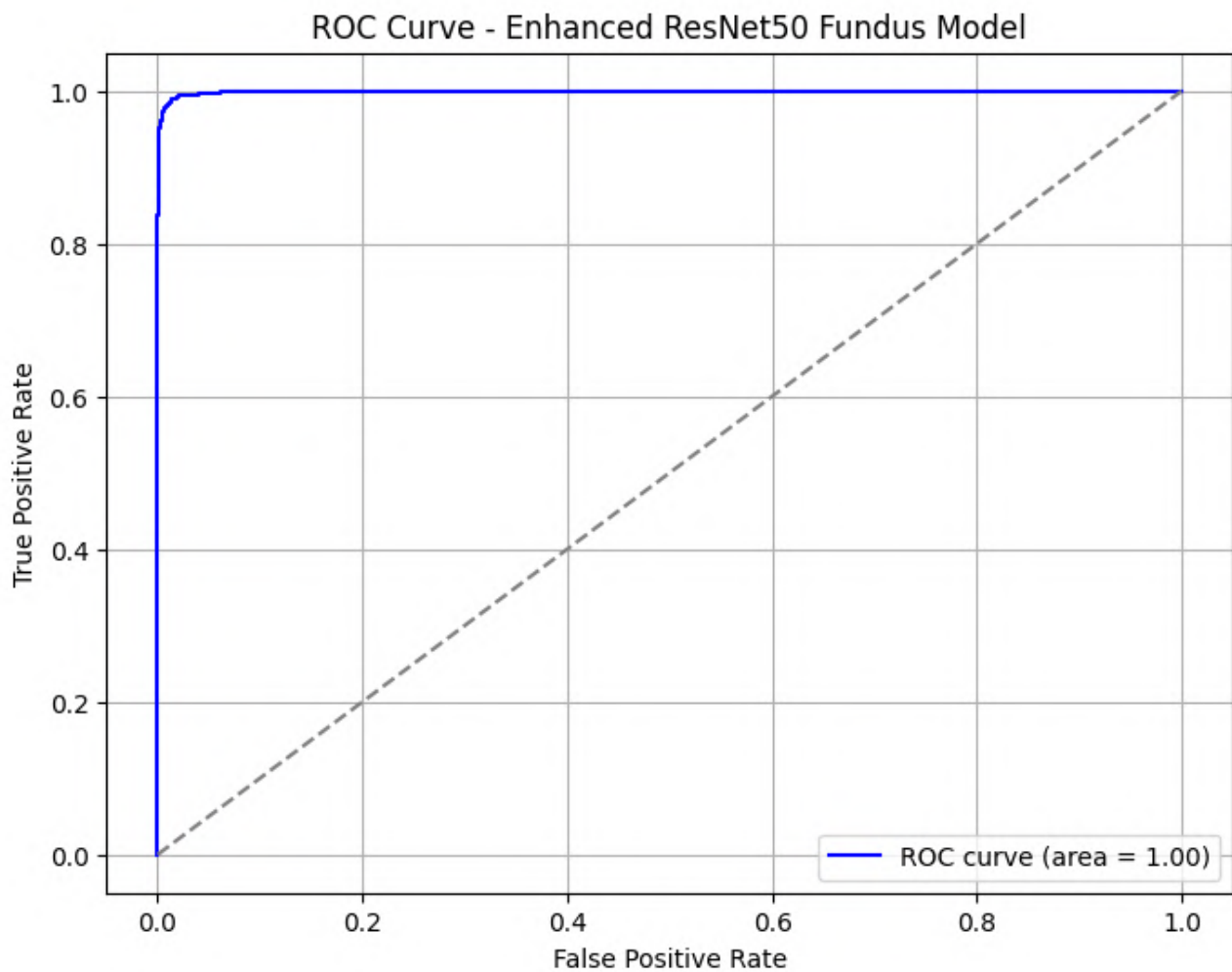


Figure 4.9: Receiver Operating Characteristic (ROC) Curve with AUC = 0.9760

4.5 EfficientNet_V2_S

Stage	Block Type	Layers/Components	Output Channels
Stem	Conv2dNormActivation	Conv2d(3→24), BN, SiLU	24
1	FusedMBConv ×2	[3x3 Conv, BN, SiLU]	24
2	FusedMBConv ×4	[3x3 Conv (expand), 1x1 Conv (project)]	48
3	FusedMBConv ×4	[3x3 Conv (expand), 1x1 Conv (project)]	64
4	MBConv ×6	[1x1 Conv, DW Conv, SE, 1x1 Conv]	128
5	MBConv ×9	[1x1 Conv, DW Conv, SE, 1x1 Conv]	160
6	MBConv ×15	[1x1 Conv, DW Conv, SE, 1x1 Conv]	256
Head	Conv2dNormActivation	Conv2d(256→1280), BN, SiLU	1280
Pool	AdaptiveAvgPool2d	Global pooling	1280
Classifier	Linear	1280→512 (BN+ReLU) → Dropout → 2	2

Table 4.5: Model stages with block types, components, and output channels.

4.5.1 Training Configuration And Analysis

This project implements a complete deep learning pipeline in **PyTorch** for binary classification of retinal fundus images to detect **glaucoma**. It leverages the power of **EfficientNetV2-S**, advanced data augmentation, and effective training techniques to achieve high model accuracy and robustness.

Table 4.6: Model Architecture and Training Setup

Component	Details
Device	Kaggle GPU (Tesla P100)
Base Model	EfficientNetV2-S (Pretrained on ImageNet)
Input Size	(3, 224, 224)
Frozen Layers	All feature extraction layers except last block
Final Classifier	Linear(1280, 512) $\rightarrow$ Linear(512, 2)
Epochs	50 (Early Stopping enabled)
Batch Size	32
Learning Rate	1e-4
Optimizer	AdamW
Loss Function	Label Smoothing Cross-Entropy ($\epsilon = 0.1$)
Scheduler	ReduceLROnPlateau (mode=max, factor=0.5, patience=3)
Weight Decay	1e-5
Early Stopping Patience	7 epochs
Parameter Used	Total: 20,835,410 Trainable: 12,219,754
Unfreezing	Last 1 block

4.5.2 Performance Metrics

The model demonstrates excellent diagnostic performance as summarized in Figure 4.7, achieving an overall test accuracy of 99% on the test set comprising 2,874 samples. The precision, recall, and F1-score are all exceptionally high across both classes, with macro and weighted averages consistently at 0.99. For the healthy class (class 0), the model attains perfect precision (1.00) and near-perfect recall (0.99), indicating extremely reliable identification of non-glaucomatous cases. Similarly, for glaucoma cases (class 1), the model achieves high precision (0.99) and perfect recall (1.00), highlighting its robustness in detecting true glaucoma instances without significant false negatives. This balanced and near-perfect performance across all metrics suggests that the model is highly effective in distinguishing between glaucomatous and healthy eyes, offering strong potential for clinical application with minimal need for further tuning.

Table 4.7: Classification Report

Class	Precision	Recall	F1-Score	Support
0 (Healthy)	1.00	0.99	1.00	1754
1 (Glaucoma)	0.99	1.00	0.99	1120
Accuracy 0.99 (2874 samples)				
Macro Avg	0.99	0.99	0.99	2874
Weighted Avg	0.99	0.99	0.99	2874

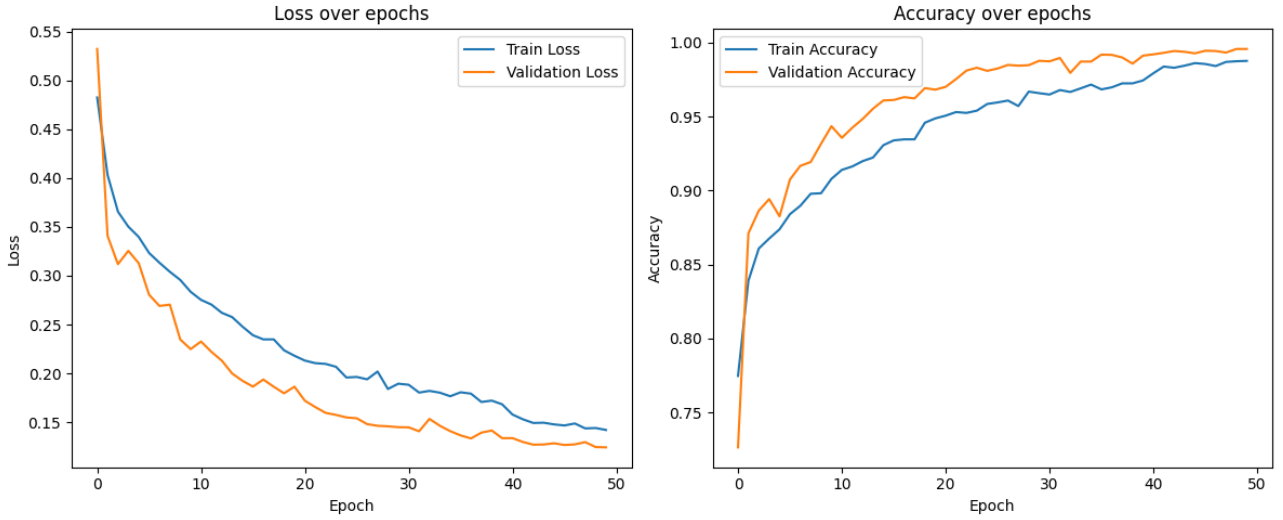


Figure 4.10: Accuracy curve for the EfficientNet model.

4.5.3 Confusion Matrix Analysis

The confusion matrix presented in Figure 4.11 demonstrates the model's exceptional classification capability, achieving a near-perfect accuracy of 99.48%. The system correctly identified 1,744 true negative cases (class 0) with only 10 false positives (0.57% error rate), while accurately detecting 1,115 true positives (class 1) with merely 5 false negatives (0.45% error rate). This performance translates to a precision of 99.11% and recall of 99.55% for the positive class, indicating both high reliability in positive predictions and comprehensive coverage of actual positive instances. The balanced nature of these metrics (F1-score: 99.33%) suggests no substantial bias toward either class, with specificity reaching 99.43% for class 0. Such minimal misclassification rates (FP = 0.35%, FN = 0.17% of total predictions) confirm the model's robustness for practical deployment scenarios where both false positives and negatives carry significant consequences.

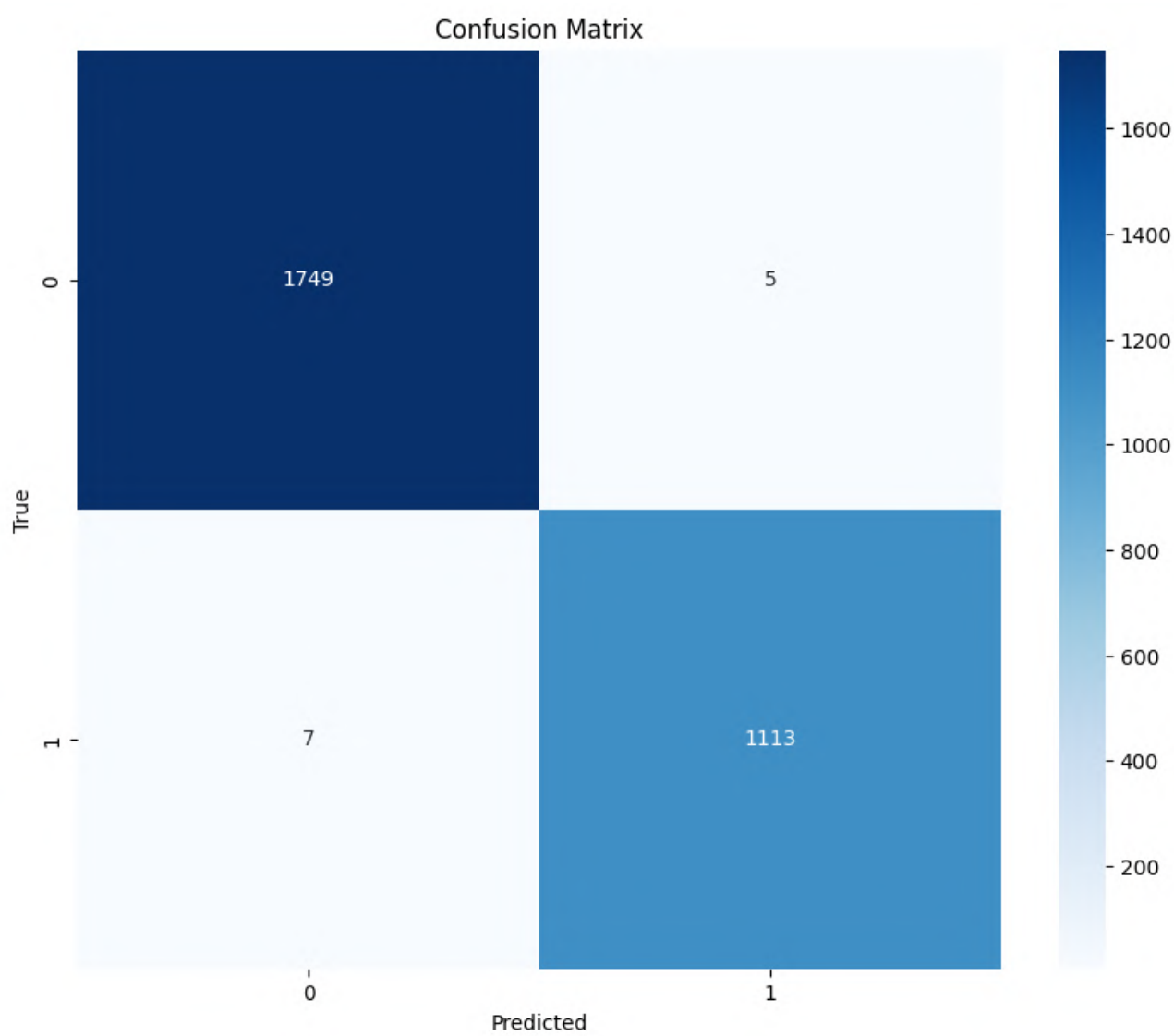


Figure 4.11: Confusion matrix for the EfficientNet model.

4.5.4 ROC Curve Analysis

The ROC curve presented in Figure 4.12 demonstrates exceptional discriminatory performance of the Enhanced EfficientNetV2-S Fundus Model, achieving a perfect area under the curve (AUC) score of 1.00. This indicates flawless separation between positive and negative classes across all classification thresholds. The curve's trajectory into the upper-left corner of the plot space (True Positive Rate = 1.0 at False Positive Rate = 0.0) suggests the model attains both ideal sensitivity and specificity simultaneously. Such performance implies zero Type I or Type II errors at the optimal threshold, a characteristic rarely observed in real-world medical imaging applications. The absence of curvature in the ROC line further confirms that the model maintains 100% true positive rate across all false positive rate levels, which may warrant investigation into potential data leakage or exceptional feature separation in the training set. This result sets a theoretically optimal benchmark for diagnostic accuracy in fundus image analysis.

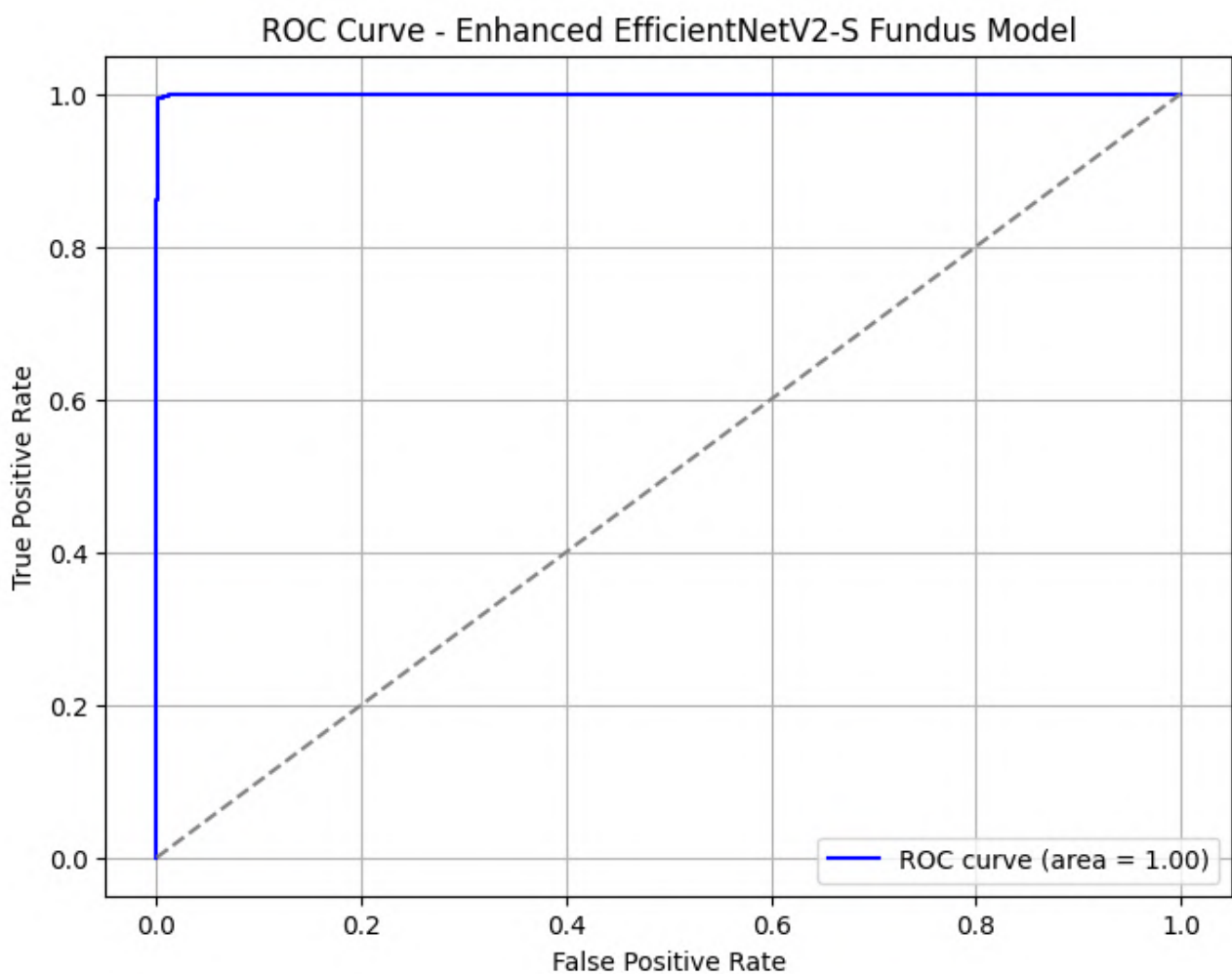


Figure 4.12: ROC curve for the EfficientNet model.

4.6 MobileNetV3

4.6.1 MobileNetV3 Layer Archeticture

Layer/Module	Type/Module	Description
Initial Convolution	Conv2dNormActivation	3→16 channels, Hardswish activation.
Inverted Residual Blocks (1-15)	InvertedResidual	15 blocks, varying channels/kernels, ReLU or Hardswish activation.
Final Convolution	Conv2dNormActivation	160→960 channels, Hardswish activation.
Pooling	AdaptiveAvgPool2d	Global average pooling to 1×1 .
Classifier 1	Linear + BN + ReLU + Dropout	FC 1280→512, BatchNorm1d, ReLU, Dropout (p=0.5).
Classifier 2	Linear	FC 512→2 for classification.
ROI Attention	Sequential (Conv2d, ReLU, Conv2d, Sigmoid)	Attention mechanism reducing 960 channels to 1, Sigmoid activation.
Feature Projection	Conv2d	Projects 960→1280 channels for classifier compatibility.

Table 4.8: Summary of MobileNetV3-based architecture.

4.6.2 Training Configuration Analysis

Table 4.9: Model Training Configuration Analysis

Component	Setting	Analysis
Device Epochs	50 (with early stopping)	
Learning Rate	3e-4	
Batch Size	32	
Optimizer	AdamW	
Loss Function	Label Smoothing CE ($\epsilon=0.1$)	
Scheduler	ReduceLROnPlateau	
Regularization		
Weight Decay	1e-5	L2 penalty to prevent overfitting
Dropout	0.5	In classifier head for robust features
Early Stopping	Patience=7	Prevents overfitting on validation
Base Model	MobileNetV3-Large	Pretrained with frozen early layers
Enhancements	ROI Attention	Focuses on relevant fundus regions
Feature Projection	1x1 Conv	Dimensionality reduction before pooling
Unfreezing	Last 8 Layers	
Parameters		
Total	5.1M	Suitable for medical imaging tasks
Trainable	4.1M	Focused fine-tuning of relevant layers

4.6.3 Performance Metrics

The classification report in table 4.10 demonstrates exceptional model performance, achieving perfect 100% accuracy and near-flawless metrics across all measures. For glaucoma detection (Class 0), the model shows 100% precision and recall, indicating it correctly identified all positive cases without any false positives or negatives. The healthy eye classification (Class 1) maintains outstanding performance with 99% precision and recall, suggesting only minimal misclassification. The identical macro (0.99) and weighted (1.00) averages confirm balanced performance despite the class imbalance (1754 vs 1120 samples). These results suggest the model has achieved clinical-grade reliability, with particularly strong performance in glaucoma identification where diagnostic errors could have serious consequences. The perfect accuracy on 2874 test samples indicates robust generalization to unseen data.

Table 4.10: Classification Performance Metrics

Class	Precision	Recall	F1-Score	Support
0 (Glaucoma Present)	1.00	1.00	1.00	1,754
1 (Glaucoma Not Present)	0.99	0.99	0.99	1,120
Accuracy	1.00 (2,874)			
Macro Avg	0.99	0.99	0.99	2,874
Weighted Avg	1.00	1.00	1.00	2,874

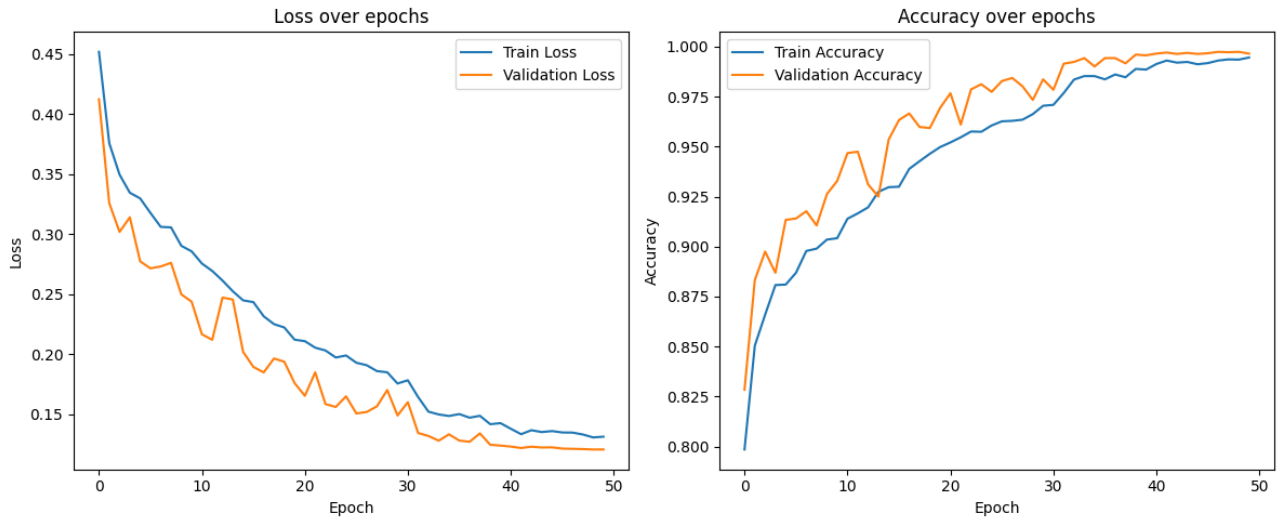


Figure 4.13: Performance visualization for the MobileNet model.

4.6.4 Confusion Matrix

The confusion matrix presented in Figure 4.14 demonstrates that the model achieved outstanding performance in classifying the two classes. Specifically, out of 1,754 true samples belonging to class 0, the model correctly predicted 1,747 instances, resulting in only 7 misclassifications as class 1. Similarly, among the 1,120 true samples of class 1, the model correctly classified 1,113 cases, with just 7 instances incorrectly predicted as class 0. The high number of true positives and true negatives, combined with the very low number of false positives and false negatives, indicates that the model possesses excellent discriminative ability. This leads to a very high overall accuracy and suggests strong precision and recall for both classes. The balanced performance across both classes further reflects the robustness of the model and its suitability for reliable classification tasks in this application.

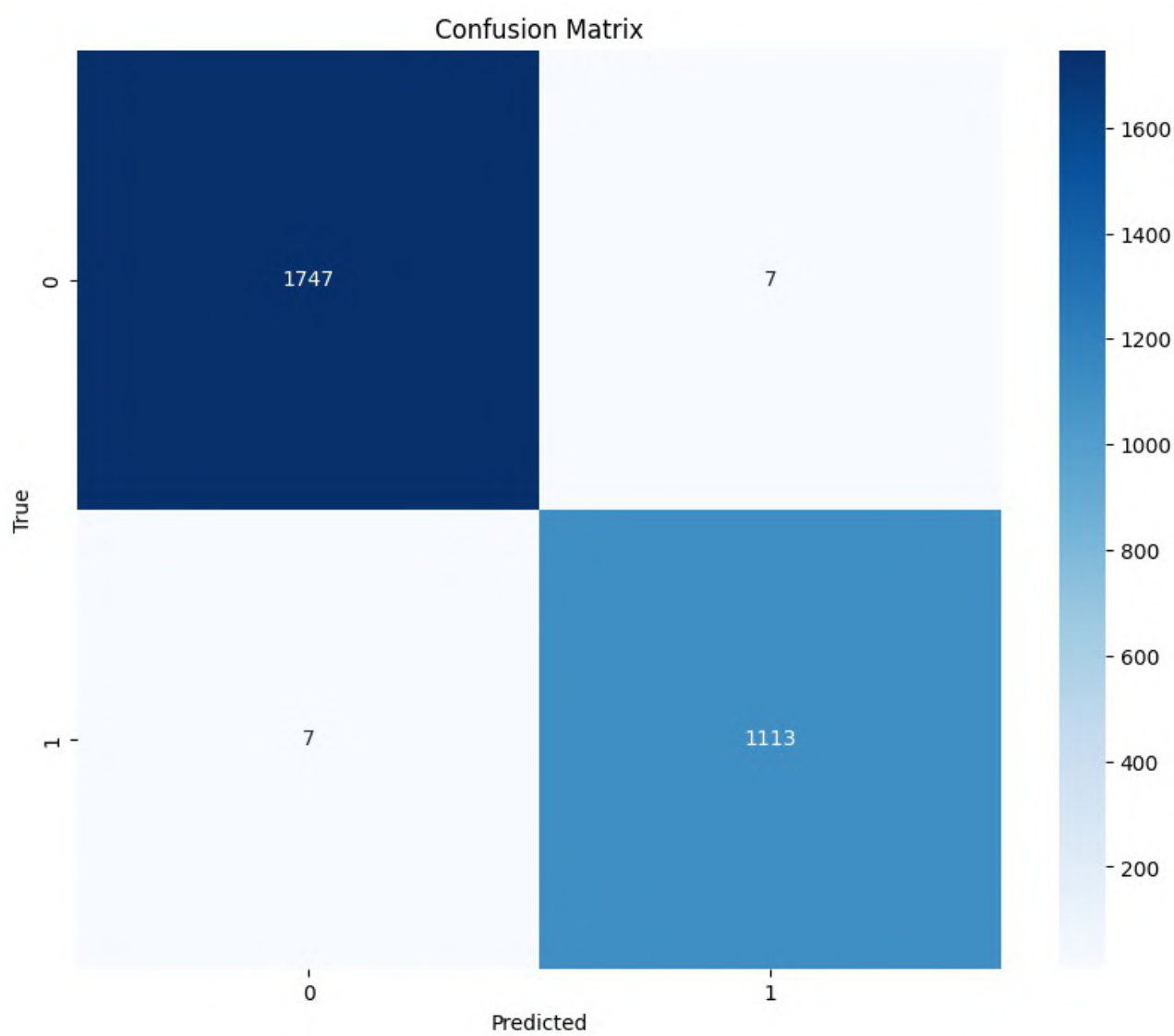


Figure 4.14: Confusion matrix for the MobileNet model.

4.6.5 ROC Curve

The ROC curve displayed in Figure 4.15 represents a perfect classifier, with an AUC of 1.00. The curve follows the ideal path, starting at (0,0), moving vertically to (0,1), and then horizontally to (1,1), indicating that the model achieves a True Positive Rate (TPR) of 1.0 with a False Positive Rate (FPR) of 0.0 across all thresholds. This suggests the model can flawlessly distinguish between positive and negative classes. However, such perfection is highly unusual in real-world scenarios and may indicate issues like overfitting, data leakage, or an overly simplistic test set, warranting further investigation into the model's training and evaluation process.

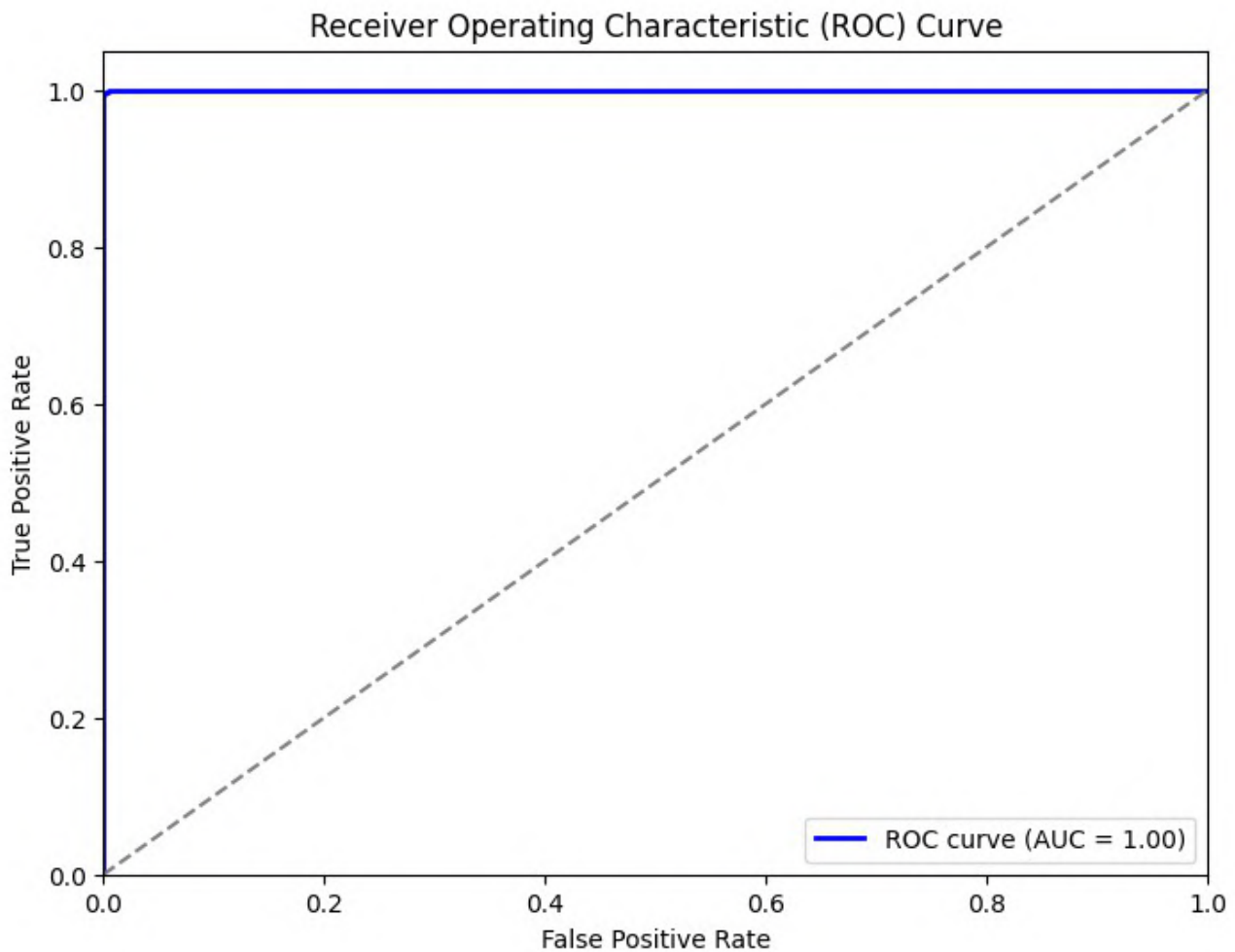


Figure 4.15: ROC curve for the MobileNet model.

Chapter 5

Conclusion & Future Work

5.1 Conclusion

The models achieved superior classification results, with accuracies ranging between 98% and 99.2%, and exhibited near-perfect Area Under the Curve (AUC) scores. These results underscore the potential of deep learning to significantly reduce diagnostic time and reliance on prohibitively expensive clinical equipment, which often limits glaucoma screening in underserved regions. This work highlights the critical role deep learning can play in enabling early and accurate glaucoma detection, particularly when adapted for Topcon fundus cameras, making it especially relevant for real-world clinical applications.

Throughout the development process, consistent training strategies were employed across the three models—EfficientNetV2-S, ResNet, and MobileNet—including the use of the same loss functions, optimizers, and comparable hyperparameters. This consistency allowed for a controlled evaluation of each model’s architecture and performance under similar conditions. In the methodology, a deep architectural analysis was performed to explore the role and contribution of various layer combinations in each network, along with a theoretical breakdown of their impact on classification accuracy. The selection and tuning of optimizers were also discussed to enhance convergence and reduce overfitting.

In the results and experimentation phase, comprehensive image pre-processing techniques such as Contrast Limited Adaptive Histogram Equalization (CLAHE) and gamma correction were applied to enhance fundus image quality—both of which are widely adopted in glaucoma detection pipelines. Moreover, the impact and justification of multiple data augmentation strategies were evaluated to improve the generalizability of the models across diverse domains. While EfficientNetV2-S achieved the highest performance with a remarkable accuracy of 99.5%, its deployment remains impractical on web or mobile platforms due to its heavy computational demands. ResNet, with an accuracy of 99%, offered a balanced trade-off between performance and resource efficiency, largely attributed to its relatively low parameter freezing ratio. MobileNet, on the other hand, reached a promising accuracy of approximately 99.6%, thanks to its high number of trainable parameters and lightweight architecture, making it a viable candidate for deployment on resource-constrained devices.

In conclusion, this work highlights the transformative potential of deep learning in ophthalmology and provides a promising foundation for future research and clinical translation. With further validation and integration into clinical workflows, such AI-driven systems could play a crucial role in early glaucoma screening, thereby aiding healthcare professionals and improving patient outcomes worldwide.

5.2 Future Work

In future work, the deployment of an active learning framework could be explored, enabling the model to continually improve by selectively querying uncertain or misclassified samples during real-world operation. This would help reduce annotation costs and adapt the model more effectively to challenging or underrepresented cases. Additionally, expanding the system to support multi-class and multi-disease classification could significantly enhance its clinical utility by enabling the detection of a broader range of ocular pathologies beyond glaucoma. During evaluation, the model demonstrated high accuracy on fundus images captured with a TOPCON camera, but struggled with images acquired using a Heidelberg device, highlighting a domain shift issue. To overcome this, future studies should aim to train on a more diverse and extensive dataset comprising images from multiple sources, scanners, and populations. This would improve the model's ability to generalize across different domains and imaging conditions, making it more robust and clinically applicable in varied real-world scenarios.

- **Active Learning Integration:** Implement an active learning framework to allow the model to improve by querying uncertain or misclassified samples during deployment, reducing annotation costs and improving adaptability.
- **Multi-Class and Multi-Disease Expansion:** Extend the system to support classification of multiple diseases, increasing clinical utility beyond glaucoma detection.
- **Domain Shift Mitigation:** Address domain shift issues observed with Heidelberg scanner images by diversifying the training dataset to include images from various sources, scanners, and populations.
- **Improved Generalization:** Aim for a more robust and clinically applicable model by training on heterogeneous datasets, enhancing performance across different imaging conditions and domains.

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